robotics

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Systematic Review

Systematic Literature Review of Vision-Language-Action Models for Generalist Robots

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Abstract

Generalist robots need to perform diverse tasks while operating in dynamic, uncertain, and unstructured environments, often around human beings. Vision-language-action (VLA) models have recently emerged as a promising and flexible framework for integrating perception, reasoning, robotic control, and action execution to develop generalist robotic policies. This systematic literature review (SLR) examines more than 140 VLA-related publications between 2020 and 2025 following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. To the best of our knowledge, it is the first PRISMA-compliant systematic review dedicated to VLA models, offering a structured discussion of robotic policies, VLA architectures, and inference optimization methods. The review also presents descriptive analyses of the included studies and a glossary defining the terminology commonly used in VLA and generalist robotic policy research. The findings reveal substantial diversity among VLA models in terms of their supported modalities, robotic embodiments, training strategies, and architectural designs. Despite the rapid growth of VLA research, several important areas remain underexplored, including the execution of complex, long-horizon tasks, effective integration of speech, and deployment on low-cost hardware, while ensuring robust, safe, and secure operation.

Keywords: vision-language-action models; VLA; robot learning; generalist robotic policy

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1. Introduction

Robots have been in use for various industrial applications since the deployment of the first industrial robot, called Unimate, in 1961. Since then, robotics as a field has made significant progress, and the current focus is on building intelligent general-purpose robots. While industrial robots are quite efficient in performing repetitive tasks in a well-structured environment, a true general-purpose robot needs to undertake diverse tasks under dynamic and complex real-world environments and therefore needs to leverage artificial intelligence-based techniques.

1.1. Background

Over the years, different machine learning approaches, such as imitation learning and reinforcement learning, have been used to develop robotic policies that enable robots to undertake complex tasks, but the successful adoption of deep neural networks, particularly transformer-based models, has revolutionized the field of robot learning. These

transformer-based models, including large language models (LLMs) and vision-language models (VLMs), have facilitated the development of a new class of models known as vision-language-action (VLA) models. VLAs have emerged as a promising approach for developing generalist robotic policies by extending the capabilities of LLMs and VLMs to action generation and robotic control. LLMs are deep neural network-based language models, often containing billions of parameters, that are pretrained on internet-scale text corpora and can subsequently be fine-tuned for specific downstream natural language processing (NLP) tasks. They have demonstrated remarkable capabilities across a wide range of NLP applications, including language understanding and generation, machine translation, question answering, sentiment analysis, and document summarization [1].

Whereas conventional large language models (LLMs) primarily process and generate text, large multimodal models (LMMs), also referred to as multimodal large language models (MLLMs), can process multiple modalities, including text, audio, images, and video. MLLMs have demonstrated strong performance in tasks such as visual question answering, image captioning, and text-to-video generation [2]. Vision-language models (VLMs) are a category of multimodal models that integrate visual and textual information to support cross-modal understanding and generation. Vision-language-action (VLA) models extend this capability by incorporating action generation, combining at least vision and language inputs to perform embodied tasks such as robotic manipulation. When a VLA architecture incorporates a pre-trained LLM or VLM, it may also be termed a large vision-language-action model (LVLA) or a large VLA [3].

VLA models are primarily developed for applications in embodied artificial intelligence, a field concerned with intelligent agents that perceive, interact with, and learn from their physical environments. As a result, these models have considerable potential across a broad range of robotics applications. VLAs typically process visual observations and natural language instructions as inputs and generate corresponding robotic actions as outputs [3]. They can produce low-level control commands for tasks such as grasping an object and transferring it from one location to another. They may also support high-level task planning by interpreting complex instructions and decomposing them into sequences of executable actions. Given the growing interest in vision-language-action (VLA) models as a promising approach to develop generalist robotic policies, there is a need to systematically examine the methods currently used to develop, fine-tune, optimize, and deploy these models. Accordingly, this systematic literature review (SLR) surveys existing approaches to VLA development and fine-tuning, together with the performance metrics used to evaluate them. The review is guided by the following research questions:

- What is the current state of the art in VLA models, including their architectures, training approaches, reported capabilities, and key strengths?
- What inference optimization methods are used to reduce VLA model size and/or improve inference speed?
- What are the main open challenges and research gaps associated with VLA models?

In order to broadly cover the research landscape in generalist robotic policies, the SLR also included questions related to different methods used to develop robotic policies and the current nomenclature used in VLA research.

Overall, this SLR offers a comprehensive examination of the current VLA landscape. It addresses fundamental questions that may help researchers and practitioners enhance their understanding, identifying research areas requiring further investigation and advancing the development of general-purpose intelligent robots capable of undertaking complex and diverse tasks in dynamic, unstructured real-world environments. While a generalist robot needs to perform locomotion, navigation, and manipulation tasks in its environment, this study is more focused on robotic manipulation.

1.2. Related Research

Recently, several notable surveys on the use of vision and language foundation models in robotics have been published, such as [4–6]. Foundation models are large models trained on internet-scale datasets that can be adapted to downstream tasks. Firoozi et al. [6] present a broad overview of foundation models and techniques used in robotics, such as language-conditioned imitation learning, language-assisted reinforcement learning, and language–image goal-conditioned value learning. It also highlights the challenges of using foundation models in robotics. Hu et al. [4] proposed a structured taxonomy of foundation models used in robotics by dividing foundation models into existing vision– language foundation models used in robotics and robotics-specific foundation models. Yao et al. [5] presented an overview of language-conditioned learning for robotic manipulation and divided them into language-conditioned reward shaping, language-conditioned policy learning, models empowered by LLMs, and lastly models empowered by VLMs. VLAs have been listed under the models empowered by the VLMs category. Firoozi et al. also [6] explored the role of foundation models in robotics. Jeong et al. [7] categorized VLAs based on how different studies use foundation models in robotics, namely for perception and situational awareness, decision-making, task planning, and motion control. A survey on dual system VLAs is presented in [8]. A dual system VLA uses two systems, namely a computationally lightweight system that generates low-level robot actions based on the output of the more complex MLLM or VLA system. The surveys in [4–7], however, do not focus specifically on VLAs. In contrast, the surveys that exclusively examine VLAs are [3,8–10].

1.3. Scientific Contributions

Given the rapid growth of vision-language-action (VLA) research and the absence of a systematically structured synthesis of the field using a method such as PRISMA, this review aims to provide a rigorous and comprehensive account of the current state of the literature. Our contributions are as follows:

- A comprehensive VLA SLR following PRISMA guidelines.
- Providing a descriptive analysis of the existing VLA literature, offering a structured overview of research trends, themes, and developments in the field.
- Identifying and summarizing the best practices for the development of VLA models, which may serve as useful guidance for both researchers and practitioners.
- Compiling the key terminology and nomenclature used in VLA research, thereby providing a reference resource for current and future researchers working on VLAs for generalist robotic manipulation.

The remainder of this review is organized as follows. Section 2 describes the methodology used to conduct the SLR, including the research questions, search strategy, and study selection criteria. It also presents the descriptive analysis of the studies included in the review. Section 3 provides a brief overview of different methods for developing generalist robotic policies. Section 4 focuses on vision-language-action models, covering their architectures, applications, evaluation and benchmarking approaches, and best practices. Section 5 discusses strategies for reducing the size of VLA models and/or improving their inference speed. Section 6 presents the current open challenges in the field. Section 7 summarizes the main results and explains how the research questions have been addressed, while Section 8 provides a general discussion of the overall findings. The limitations of this review are presented in Section 9, followed by the conclusions in Section 10.

2. Research Methodology

To understand the current VLA research landscape, we conducted a systematic literature review of research publications from January 2020 to August 2025, drawing on several databases. The methodology, based on the PRISMA guidelines for systematic reviews, was structured into four phases: research planning; literature identification and screening; data extraction and analysis; and, finally, interpretation and reporting. The overall research methodology is illustrated in Figure 1, while the complete PRISMA checklist is provided in Supplementary Materials Table S1.

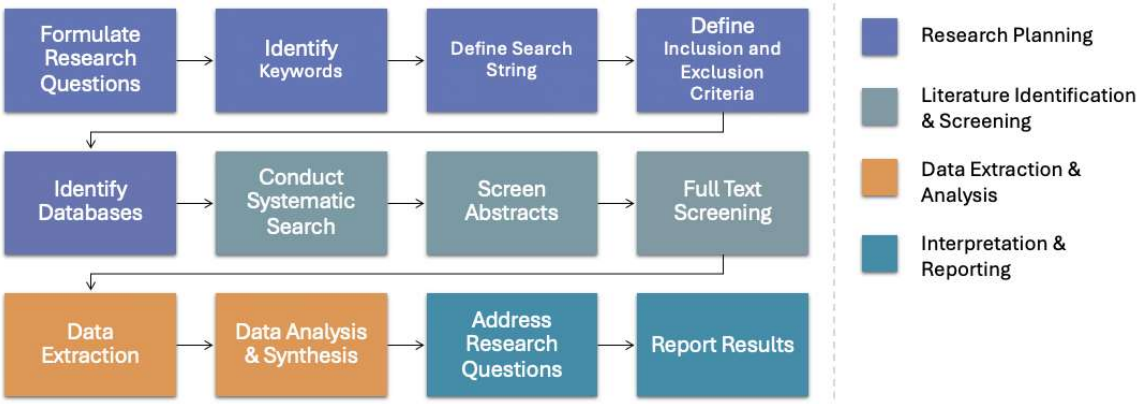


Figure 1. The systematic literature review methodology.

2.1. Research Planning

This phase focused on formulating the research questions and defining the search strategy.

2.1.1. Research Questions

The planning phase started with framing the following research questions:

- RQ1: What are the different methods used to develop robotic policies in addition to VLAs?
- RQ2: What is the main nomenclature used in VLA and generalist robotic policy research?
- RQ3: What is the current state of the art in VLAs in terms of architecture and training approaches?
- RQ4: What are the main inference optimization methods used to reduce the size and/or increase the inference speed of VLAs?
- RQ5: What are the current open challenges in VLAs?

RQ1 aims to examine methods that precede VLAs or are used in tandem with VLAs to develop robotic policies. This is important to understand how different artificial intelligence-based approaches have evolved over time, from specialist single-task robotic policies to VLA-based generalist robotic policies.

RQ2 seeks to discover the nomenclature used in VLAs and generalist robotic policy-related research. This is intended to help current and future researchers and practitioners in robotics develop a high-level understanding of commonly used terms in the existing generalist robotic policy-related literature.

RQ3 investigates the current state of the art in VLAs, their architecture, and how they are trained. There has been a strong interest in the research community to enhance robustness, generalization, inference speed, and other performance metrics of VLAs. This question is meant to compare the differences in architectures and training approaches and, consequently, the unique strengths of the VLAs covered in this study.

RQ4 explores the inference optimization techniques being used to balance the performance between model size, inference speed, and accuracy. This is one of the most important topics in VLA research, as larger model sizes can make it difficult to deploy these models on robots with limited compute capacity and can have a negative impact on the performance of the robot during the inference stage in practical applications. On the other hand, smaller models may fail to offer the kind of generalization that can help create a generalist robot policy to handle diverse tasks under different environments. In many ways, the goal of coming up with an efficient VLA is to develop a model with sufficient generalization capability to handle diverse tasks, reasonable inference speed to be usable in practical applications, and a model size that is easy to deploy on edge devices.

RQ5 identifies current open challenges in VLAs, including safety and security issues. While existing surveys have considered aspects of RQ3, RQ4, and RQ5, the rapid evolution of the VLA landscape warrants a renewed examination of these questions in this SLR.

2.1.2. Search Strategy

The keywords related to the research were identified as follows based on their relevance to the research questions, with particular emphasis on VLAs and generalist robotic policies.:

KW1: Vision.
KW2: Language.
KW3: Action.
KW4: VLA.
KW4: Robot.
KW5: Imitation.
KW5: Learning.

During the initial search, language-conditioned imitation learning was also included to enable comparisons with VLAs. Accordingly, the keyword conjunctions used in the advanced search for different databases included one or more of the following combinations: (vision AND language AND action), (vision AND language AND action AND robot), (VLA AND robot), (imitation AND language AND learning), and (imitation AND learning AND language AND robot). The search strategy was adapted for each database to account for differences in indexing practices, searchable fields, controlled vocabulary, and query syntax. Although the exact keyword conjunctions varied to increase the breadth of coverage, all searches were designed to capture the same core concepts relevant to the review question, and the complete database-specific strategies are reported in Supplementary Materials Table S2. In addition to these conjunctions, the advanced search feature of different databases was used to limit the search to manuscripts published in the English language between 2020 and 2025, and they had to be related to math, engineering, or computer science where possible. The initial search was conducted in March 2025 and was updated on 13 August 2025 through a free search and therefore does not contain publications during the remaining period of 2025.

It is important to mention that different databases offer different kinds of filters and advanced query syntax, so it may not be possible to use the same filters in all cases.

As an example, the free search using Google Scholar did not allow searching a keyword or conjunction in the abstract at the time of conducting the search, so the keywords were searched only in the title using a syntax like ‘allintitle: vision language robot’. This relaxed conjunction by removing the word ‘action’ allowed getting more hits, as the search could not be done in the abstract. A graphical user interface was used for databases where advanced search queries could not be incorporated.

The inclusion and exclusion criteria to decide which studies are relevant and should be included in the SLR are given in Table 1, and the databases used to search for the relevant studies are given in Table 2.

Table 1. Inclusion and exclusion criteria.

Inclusions	Exclusions
IC1: The title or abstract of the publication includes the selected keywords	EXC1: Papers for which the full text is not accessible
IC2: Papers published in the proceedings of a conference, journal, or with a preprint available at arXiv	EXC2: Papers that are about navigation (visual language navigation)
IC3: Available in one of the selected databases	EXC3: Papers that are related to robotics but without using large language models
IC4: Published within the last five years (2020–2025)	EXC4: Papers that focus on models for action anticipation
IC5: Mentions vision-language-action models and robotics	EXC5: Papers that do not use a visual modality as input
IC6: Publication is related to engineering, mathematics, or computer science	

Table 2. Selected databases and digital libraries used for searching relevant studies.

Database	Type	URL
ACM Digital	Digital Library	https://dl.acm.org
arXiv	Digital Open Access Archive	https://arxiv.org
Google Scholar	Search Engine	https://scholar.google.com
IEEE Xplore	Digital Library	https://ieeexplore.ieee.org/Xplore/home.jsp
ScienceDirect	Digital Library	https://www.sciencedirect.com
Scopus	Digital Library	https://www.scopus.com/home.uri
Semantic Scholar	Digital Library	https://www.semanticscholar.org/me/research
Springer	Digital Library	https://www.springer.com/gp

2.2. Literature Identification and Screening

The identification and screening process followed the PRISMA guideline, with the overall flow given in Figure 2. During identification, 1722 studies were found through database and free-text search, and 465 duplicates were removed. In all, 1257 studies were screened using the title and abstract to remove irrelevant studies as per the inclusion and exclusion criteria given in Table 1. The final list of 142 studies was reviewed in detail to answer the research questions formulated for this study. The list of included studies is reported in Supplementary Materials Table S3.

The initial title/abstract screening and full-text eligibility assessment were conducted by one reviewer using the predefined inclusion and exclusion criteria. Therefore, independent duplicate screening was not performed, and no formal inter-rater agreement statistic was calculated. To reduce the risk of selection or extraction error, the final set of included studies, extracted variables, evidence tables, synthesis, and conclusions were reviewed by all authors through iterative discussion and consensus.

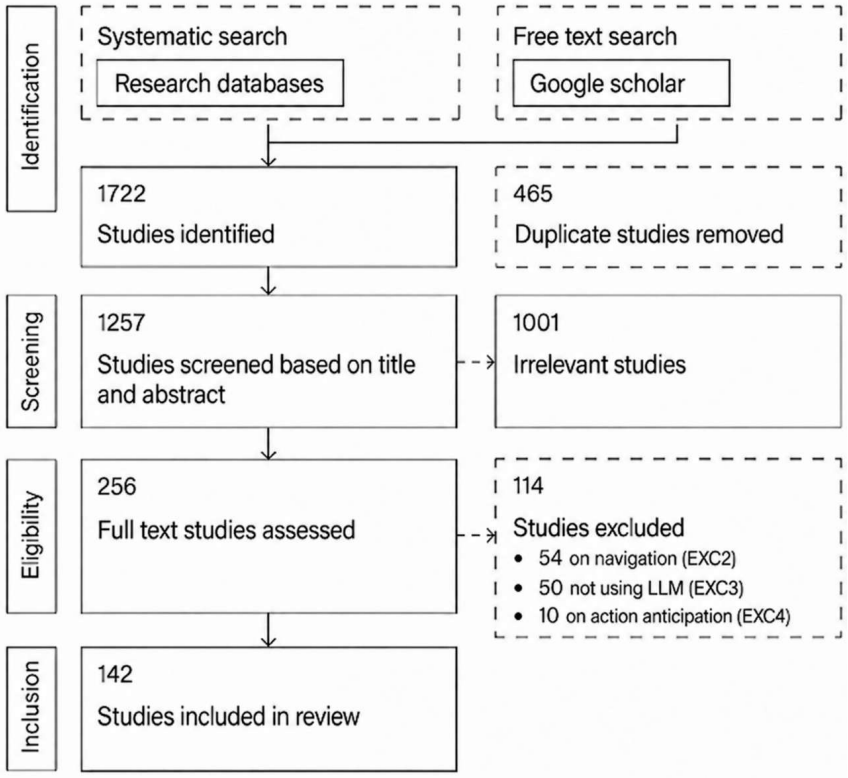


Figure 2. PRISMA flow for systematic review.

It is important to mention that in this SLR, any studies related to visual language navigation (VLN) models that enable agents to navigate in a physical or simulated environment in response to a natural language instruction were excluded, as VLAs focus more on robotic manipulation tasks. It is important to mention that VLNs are like VLAs to the extent that both use human language instructions and visual modality at the input to undertake a given task, but VLNs are mostly for mobile robot navigation, whereas VLAs mainly focus on manipulation tasks.

2.3. Quality Assessment, Data Extraction, and Analysis

In this phase, data were extracted from each study included in the systematic literature review and analyzed to answer the research questions and assess the maturity of research on vision-language-action models and generalist robotic policies. The data extraction process began by exporting publication metadata from Zotero version 7.0.13 [11].

The exported comma-separated value (CSV) file contained several fields, of which the following were retained for analysis: Key, Item Type, Publication Year, Author, Title, Abstract Note, and File Attachments. Full texts were extracted from PDF attachments using a Python v3.11 script.

The titles and abstracts of 1257 records were screened against the predefined inclusion and exclusion criteria. A total of 256 studies were then assessed in greater detail for eligibility, resulting in a final set of 142 included studies.

For each of the 142 included studies, the following information was systematically extracted: item type, architecture description, model name, model size, model inference speed, robot platform, success rate for seen and unseen tasks, evaluation metrics, datasets, benchmarks, optimization methods, reported challenges, and future directions. Where information was not reported in a study, the corresponding field was left blank rather than inferred.

This review was not prospectively registered in the International Prospective Register of Systematic Reviews (PROSPERO). The certainty of the evidence was not formally assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework, and clinical risk-of-bias assessment tools were not applied. These approaches were considered unsuitable for the included robotics and artificial intelligence studies because of their substantial heterogeneity. Instead, the review reports descriptive patterns across the literature, including publication trends, model types, datasets, benchmarks, robot embodiments, availability of code, and reported performance metrics. Although quality assessment was not considered in the initial review plan, a post hoc quality assessment was conducted to evaluate the methodological rigor of the included studies and strengthen the interpretation of the findings. Because this review focuses on heterogeneous robotics and AI studies rather than clinical interventions, a conventional clinical risk-of-bias tool was not directly applicable. Instead, the quality assessment considered whether each study clearly reported its model architecture, datasets, training details, evaluation design, generalization testing results in terms of seen and unseen task performance where applicable, robustness testing, reproducibility potential, and stated limitations or failure cases. Public availability of code for the included studies as reported in Supplementary Materials Table S4 was used as a key criterion to assess reproducibility potential for individual studies.

This post hoc quality assessment mainly evaluates methodological quality, reporting transparency, and reproducibility rather than a typical risk-of-bias assessment. The quality appraisal scoring criteria are provided in Supplementary Materials Table S5, and scores assigned to each study are given in Supplementary Materials Table S6. All scores are numerical, other than for reproducibility potential, where code availability, model architecture description, dataset details and level of training details were used to classify each study as having low, moderate or high reproducibility potential. If a particular quality criterion is not applicable to the study, 'N/A' was assigned instead of leaving it blank.

Overall, only 64% of studies included in the review shared code through public repositories. Table 3 summarizes the post hoc quality assessment based on reporting adequacy across the included studies. The (n/N) field represents the number of studies meeting a given quality assessment criterion (n) relative to the total number of studies for which that criterion was applicable (N), excluding studies marked as not applicable (N/A). Model architecture was the most consistently reported criterion, with 92% of studies providing sufficient architectural information, followed by training details at 75%. Evidence concerning generalization was adequately reported in 60% of the studies. Reproducibility potential was high in slightly more than half of the studies (52%). In contrast, fewer than half adequately evaluated robustness (47%) or explicitly reported study limitations (48%).

Reporting bias was considered qualitatively. The review may be affected by publication and availability bias because studies with positive benchmark results, publicly released models, or high-profile preprints may be more visible and more likely to be indexed or cited. To reduce this risk, both database searches and free-text searches were used, and preprints were included if they met the eligibility criteria. However, the inclusion of preprints also introduces uncertainty because some included studies may not have undergone peer review at the time of inclusion.

The extracted data were tabulated for structured comparative analysis. Descriptive metrics, including the number of publications per year, country of origin, frequency of datasets, benchmarks, robots, and embodiments, were aggregated and plotted for comparison. If a study cited another work with a clear emphasis on generalist robotic policies, the cited study was skimmed to identify and extract any relevant information, where available, to support comprehensive coverage of the topic.

Table 3. Post hoc quality assessment.

Quality-Assessment Criterion	(n/N)	Studies Meeting the Criterion (%)
Model Architecture Description	130/142	92
Training Details	91/122	75
Generalization	84/139	60
Robustness	65/139	47
Reproducibility Potential	74/142	52
Limitations	68/142	48

A citation network visualization was also developed using citation graph generator to show citation relationships among the studies included in the review. During analysis, newly encountered terminology was searched and added to a separate terminology table together with the reference from which it was sourced.

2.4. Interpretation and Reporting

In this phase, the analyzed data were used to find relevance to the research questions, with the aim of answering the questions formulated in this study. The findings were collated and presented in this SLR during this final phase.

2.5. Descriptive Analysis

During data extraction and analysis, descriptive analysis was performed on the selected studies. As part of the descriptive analysis, the geographic distribution analysis revealed that most of the research in VLAs was carried out by researchers in the United States and China. It was also determined that the number of publications related to VLAs has been increasing every year since 2022. The yearly frequency of publications and geographic distribution of VLA researchers are given in Figure 3.

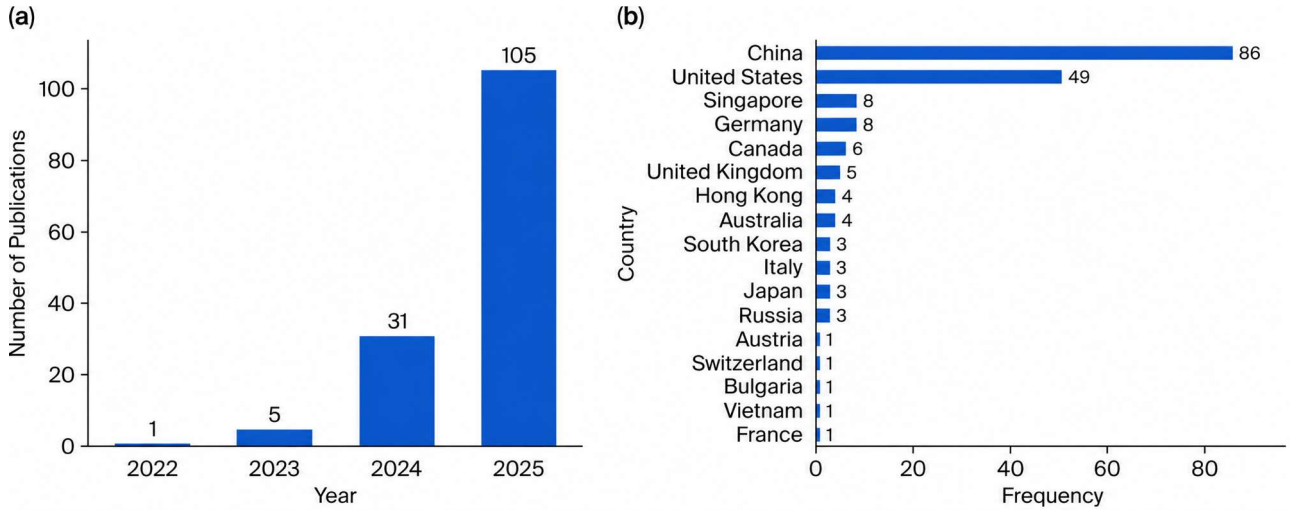


Figure 3. Descriptive analysis: (**a**) number of publications (2022–August 2025) and (**b**) researchers' country of origin.

3. Generalist Robotic Policies

This section provides essential background for understanding the mathematical formulation of generalist robotic policies and the key terminology used in the VLA literature, along with the evolution of robot learning that eventually led to the development of VLAs.

3.1. Generalist Robot Policy for Multimodal Models

In terms of multimodal models for robotics, for any action that a robot takes, we can consider output y as a function of multisensory input x and context c, as expressed in equation form by Hu et al. [4]:

$$\mathrm{y_{tk}} = f(\mathrm{x_{tk}}, c_k); \forall \mathrm{k} \in \mathrm{N}, \forall \mathrm{t} \in \mathrm{T}. \tag{1}$$

In other words, for a given task k, at a given timestep t from an episode consisting of a set of timesteps T, an action taken by the robot is a function of information from different sensors and the context provided by the language or any other modality. For VLAs, the main sensor is usually one or more cameras but can also include other sensors.

A generic conditioned stochastic robotic policy π with parameter θ that maps robot state and observations to an action distribution can be described via Equation (2):

$$\pi(\mathrm{a}|\mathrm{s}, \mathrm{o};\ \theta) \tag{2}$$

where **s** is the robot state (e.g., joint angles, pose, velocities, gripper state, etc.), **o** is the observation of the environment using one or more sensors (e.g., image, video, depth, thermal infrared, point cloud, etc.), and **a** is the action (e.g., motor commands, pick, drop, move, etc.). When the output is an action vector, it consists of joint angle or position deltas and the gripper state for low-level control policies. It is, however, important to note that for VLAs that act as action-tokenized autoregressive next-token prediction models with greedy decoding at the inference stage, the action output is deterministic.

For a VLA generalist robot, one possible formulation of a multi-task stochastic policy that outputs a probability distribution over actions for a given task k at timestep t is given in Equation (3):

$$\pi(\mathrm{a_{tk}}|\mathrm{V_{tk}}, \mathrm{L_k}, \mathrm{s_{tk}}; \theta); \forall \mathrm{k} \in \mathrm{N}, \forall \mathrm{t} \in \mathrm{T} \tag{3}$$

where

$\mathrm{V_{tk}}$ = visual observation or observations through one or more cameras at time t.

$\mathrm{L_k}$ = language instruction for task k.

$\mathrm{s_{tk}}$ = robot state at time t.

$\mathrm{a_{tk}}$ = action or action vector for task k at time t.

T = set of timesteps that constitute the trajectory needed to complete task k.

It is important to mention here that the formulation in Equation (3) is only for the case of VLAs being used as low-level control policies, as VLAs can also undertake high-level task planning. The same multi-task generalist policy, when incorporating history comprising states and observations from previous timesteps, is given in Equation (4):

$$\pi(\mathrm{a_{tk}}|\mathrm{V_{\tau k:tk}}, \mathrm{L_k}, \mathrm{s_{\tau:tk}}; \theta); \forall \mathrm{k} \in \mathrm{N}, \forall \mathrm{t} \in \mathrm{T} \tag{4}$$

where τ represents the number of timesteps before the current timestep. If the policy is only considering the current state along with the last state, then $\boldsymbol{\tau}$ starts from $\mathbf{t} - \mathbf{1}$. For a history of **n** previous timesteps, $\boldsymbol{\tau}$ starts from $\mathbf{t} - \mathbf{n}$, where **t** represents the current timestamp.

3.2. Relevant Terminology

VLA models are one of many artificial intelligence-based methods used in robotics. To understand the literature related to VLAs and other relevant robot learning fields, it is important for researchers to develop a good understanding of some of the key terminology used in this field, and it has been added as a glossary in Appendix A.

3.3. Robotic Policy Learning Methods

The following are some of the main techniques used to devise robotic policies:

- Imitation learning (IL).
- Reinforcement learning (RL).
- Meta-learning (ML).
- Diffusion policy (DP).
- Transformer-based models (e.g., LLM, VLM, VLA, RFM, etc.).
- Hybrid learning (HL).

In imitation learning, an agent learns to perform tasks by observing expert demonstrations of that task. In [12], as an example, imitation learning was used to devise a conditional policy that used RGB images and natural language task commands or a video of a person to generate actions for a 7-DOF robot at 10 Hz. This approach gave an average success rate of 44% on new, unseen manipulation tasks. Behavior cloning (BC) is used as one of the most common approaches in imitation learning to clone the demonstrated expert behavior. While this requires costly task segmentation and labeling, another approach that allows reasonable generalization capability is to use unlabeled play data, as in [13]. In general, the limitations of imitation learning are that the model cannot exceed the performance of the demonstrator and, secondly, cannot generalize to tasks outside the demonstration data [14].

In reinforcement learning (RL), the agent learns a policy through trial and error by interacting with its environment and receiving a scalar reward signal. The objective is to maximize the expected cumulative reward over a trajectory of actions. Deep reinforcement learning is used to develop MT-Opt, a scalable and generalizable multi-task learning method that can undertake new tasks that are different from the tasks in the training dataset [15]. While RL has been found to be quite effective for learning single-task policies, pure RL, i.e., without combining it with other methods, is not promising for creating a multi-task policy.

Meta-learning, often described as 'learning to learn,' can facilitate generalization by enabling robots to undertake new tasks in different environments by using limited training data. For instance, a robot can learn to undertake a task using a single video of a human performing the same task [16].

Diffusion models are a class of generative artificial intelligence (GAI) models that were originally developed for image synthesis. The models are trained using forward diffusion, which adds noise to a sample, and reverse diffusion, which removes noise from a sample. Diffusion policy [17] has denoising diffusion probabilistic models (DDPMs) at its core, where a denoising process creates the output. Diffusion-policy-based visuomotor control policies have been found to be more effective than pure BC-based methods, as per the comparison given in [17].

Transformer architecture, since its inception in [18], has revolutionized language modeling and resulted in the development of large language models, large multimodal models, robotic foundation models, VLAs, etc. Transformer-based models have been used in robotics for high-level task planning and reasoning [19], scene understanding, and low-level control of robots [7]. EmBERT [20] was one of the earliest attempts to use a transformer-based model for language-conditioned task completion in robotics, and since

then, a wide range of very promising techniques to develop generalist robots have been devised using transformer-based models, including the VLA family.

Hybrid learning refers to the use of two or more learning techniques together to create a generalist robotic policy. SayCan combines an LLM with RL to learn a language-conditioned value function that enables the robot to execute instructions by selecting actions it can successfully complete according to an affordance function [21]. Inverse reinforcement learning has been used to learn a reward function from videos of experts carrying out the same task under different embodiments [22]. This policy could then be scaled to other agents. An example of a task-conditioned multi-task policy is presented in [23]. A multi-task, multi-embodiment, multimodal generalist policy is proposed as Gato in [24], where data from diverse modalities such as text, images, proprioception, etc., are used to train a transformer sequence model. Gato's evaluation of real robot skill generalization has shown an average success rate of 50.2%. VIMA [25] is a robot agent that leverages a multi-task encoder–decoder architecture and behavioral cloning to train the first model that could accept multimodal input prompts.

The Octo model [26] is a transformer-based diffusion policy that works out of the box with different robots and can be fine-tuned with respect to new robot setups. It can also be fine-tuned to new inputs and action spaces. Another example of a hybrid learning algorithm is action chunking with transformers (ACT), which is a transformer-based imitation learning algorithm that enables robots to learn to undertake difficult tasks with an 80–90% success rate using only 10 min of demonstrations [27]. The most recent example of a hybrid model is SRT-H, a hierarchical framework for language-conditioned imitation learning that enabled a robot to perform gallbladder removal surgery with minimal human intervention [28].

Similarly, VLAs have been fine-tuned using offline and online reinforcement learning, such as ConRFT, where this fine-tuning approach has led to a significant increase in average success rates for the tasks that the model has been fine-tuned on [29].

3.4. Evolution of Language- and Vision-Based Robot Learning

Using natural language to control robots is not entirely a new research topic. For example, in [30], a system was proposed that takes commands given in the English language and the background context information and translates them into instructions for robotic actions. In [31], a working prototype developed in LISP demonstrated how English sentences are translated into robot instructions using context-free grammar and parsing. In [32], the authors shared the research carried out in the field of instructible robots by looking at instructions for mobile robots, robotic arms, etc. Self-organizing maps (SOM)-based hierarchical models were used for robotic control, leveraging the concept of regional brain modularity with multiple SOMs, each performing one part of the overall task assigned to the robot [33]. In [34], researchers suggested a cell-assembly-based model that uses visual, language, motor, and planning parts to enable a robot to understand spoken commands. A hierarchical grounded situation model (GSM) combining linguistic, proprioceptive, and visual information is recommended by [35]. An imitation-learning-based approach was proposed in [36], where a student model takes multimodal input of vision, language, and motor command modalities to mimic the actions of a teacher using Kohonen self-organizing networks and Helmholtz machines. It is noteworthy that the significance of combining language, vision, and action predates the LLM and VLA emergence, as [37] developed an architecture of a system that combines these modalities for robot control, while [38] highlighted the significance of integrating language and action inspired by the tight coupling of language and action processing in the human brain. Another framework based on action and object hierarchy to interpret natural human language commands, conceptual-level

reasoning, and executable actions was formulated by [39]. After the emergence of LLMs, frameworks such as ProgPrompt [40] showed how LLMs can be used to generate and execute robotic task plans.

After the phenomenal success of LLMs, the research moved into combining different modalities, and some of the research provided the foundation for subsequent work in VLAs, such as the development of contrastive language–image pre-training (CLIP) [41]—which combines image and text modalities and was later used in robotics, such as the research in [42]—where the system called LM-Nav takes images from the target environment and free-form language-based instructions to control the navigation of a robot. CLIPort, in [43], is a language-conditioned imitation learning agent for manipulation tasks that uses a semantic and spatial stream-based architecture based on CLIP and Transporter ResNets, respectively, for tabletop manipulation tasks.

Florence is a foundation vision model that can be adapted for downstream vision-language tasks such as video reasoning and visual question answering (VQA) [44]. One of the earliest studies in the development of multimodal, multi-task, and multi-embodiment generalist agents was Gato [24], proposed by Google DeepMind. PaLM-E is a generalist embodied language model where multimodal information is injected into a pre-trained LLM's embedding space to improve robot planning tasks. RT-1 is a transformer-based model that is trained using imitation learning, which takes images and natural language tasks as inputs. RT-1 showed a 97% success rate on seen tasks and a 76% success rate on unseen tasks [14]. RT-1 uses a universal sentence encoder to encode instructions into embedding vectors that are fused with a pre-trained EfficientNet-B3 vision model using feature-wise linear modulation (FiLM) [14]. One of the most important steps of robotic manipulation is grasping the target object, especially in a cluttered environment. Xu et al. [45] were one of the first to use joint modeling of vision, language, and action for grasping target objects.

ManipLLM [46] used CLIP as a visual encoder and LLaMA as a text encoder and kept them frozen while fine-tuning them to predict poses for robotic manipulation.

RoboCat [47] is a multi-task generalist agent for robotic manipulation that uses an architecture similar to Gato. RoboCat is reported to yield good generalization capabilities if fine-tuned on a small number of demonstrations for new tasks or new embodiments. The first well-known VLA model used for robotics, however, was RT-2.

RT-2 was developed by directly training vision–language models for converting visual question answering into low-level robot actions [48]. To develop RT-2, vision–language models are co-fine-tuned on vision–language tasks and robotic trajectory data. RT-X [49] models trained on the Open X-embodiment dataset were found to have a higher success rate and better generalization capability.

Another model called RT-H is based on creating an action hierarchy where a task assigned through natural language is translated into an intermediate language motion, which subsequently is converted into a robot action [50]. OpenVLA [51] was the first open-source vision–language–action model that later became one of the most common references for comparing the performance of other VLAs developed later.

4. Vision-Language-Action Models

Vision-language-action (VLA) models can be defined as models that take multimodal inputs, including visual and language information, and produce robotic actions as outputs [3]. In general, VLA models are meant to integrate visual perception and semantic understanding to undertake some embodied task.

They can be used for both high-level robot task planning and low-level robotic control. VLAs may leverage existing pretrained vision–language models and adapt them through

fine-tuning or co-fine-tuning; depending on the architecture, actions may be represented as discrete action tokens or generated as continuous action chunks.

Common components of VLAs include a vision encoder that converts one or more RGB images or RGB-D images into vision tokens, a language tokenizer and embedding layer that converts textual instruction into language tokens and corresponding embeddings, a pre-trained LLM or VLM, and an action head, decoder or action expert depending on the architecture. An example of VLA architecture is given in Figure 4.

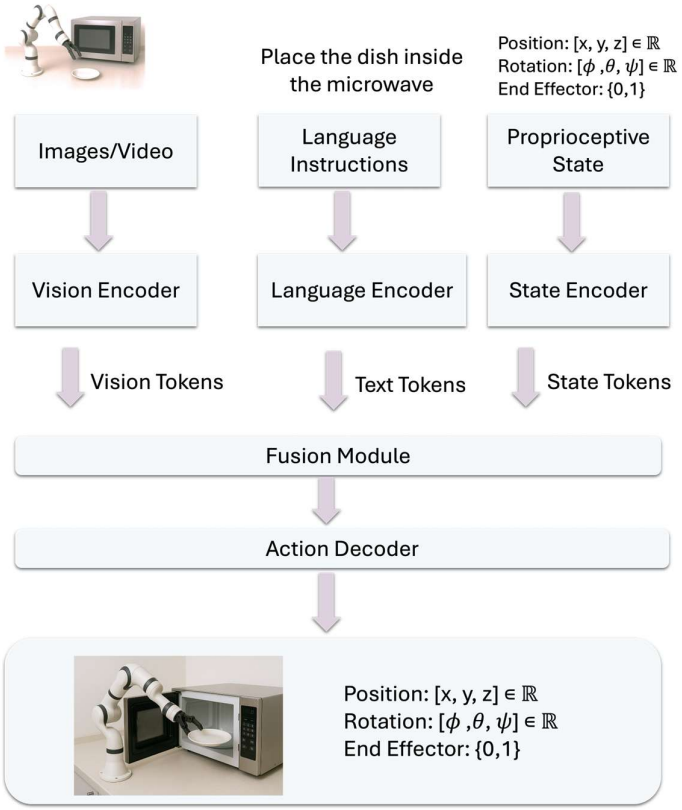


Figure 4. Example of VLA architecture.

In terms of how VLAs commonly work at the inference stage, one or more images are encoded into vision token embeddings; a language-based instruction about the task or goal for the robot is encoded into text embeddings; and optionally, the robot's proprioceptive state information from its sensors, such as joint positions or angles, joint velocities, and gripper or end-effector state, is encoded by a state encoder or projection mechanism. The resulting visual and language representations, together with proprioceptive state information when supported, condition the model's action-prediction mechanism. Depending on the architecture, the VLM/LLM backbone may directly predict discrete action tokens, or its representations may condition a separate action head, decoder, or action expert that generates one or more continuous robot actions. The predicted actions are ultimately represented or decoded as one or more numerical action vectors. Action vectors may encode end-effector poses or pose deltas, joint-space positions or deltas, velocities, gripper commands, or other embodiment-specific control quantities. The end-effector pose vector or pose deltas can be given in translational coordinates (x, y, z) and orientations, represented, for example, using roll, pitch, and yaw (ϕ, θ, ψ), respectively. Gripper control may be represented as a binary or discrete open/close command, or as a continuous target gripper position when finer control is required. While this describes a common VLA design pattern, there is substantial variation in VLA architectures and in the kinds of inputs and outputs they support.

4.1. State-of-the-Art VLA Models

The successful demonstration of early vision-language-action (VLA) models, such as RT-1, RT-2, and OpenVLA, in learning generalist robotic policies has stimulated considerable interest in developing more effective and computationally efficient VLA architectures. Subsequent research has explored variations in model architecture, training data, action representation, robotic embodiment, and other design choices. Current state-of-the-art VLA models exhibit substantial diversity in their vision and language encoders, action-decoding mechanisms, training datasets, supported embodiments, and optimization strategies. Table 4 summarizes several of the principal design variations observed across these models.

A vision encoder processes one or more input images and transforms them into visual tokens or feature representations. Commonly used vision encoders in contemporary VLA models include contrastive language–image pre-training (CLIP), vision transformers (ViTs), sigmoid loss for language–image pre-training (SigLIP), and DINOv2, which is also based on the ViT architecture. The ViT introduced in [52] adapts the transformer architecture to visual inputs. Rather than relying on conventional convolutional neural networks, it divides an RGB image into fixed-size patches, such as 16 × 16-pixel regions, and projects these patches into embeddings that can be processed by transformer layers. In robotic perception, these representations may support tasks such as object recognition, scene understanding, and manipulation. CLIP is a pre-trained multimodal model that maps images and textual descriptions into a shared embedding space [41]. SigLIP [53] adopts a similar image–text representation-learning approach but replaces the softmax-based contrastive objective with a sigmoid loss, which can improve computational and memory efficiency during training. Language or text encoders are typically transformer-based models that convert natural-language instructions into embeddings, enabling the VLA model to interpret task descriptions and condition its actions on the provided commands.

Table 4. Variations in VLAs.

Variation Type	Variation Examples
Vision Encoder	CLIP, ViT, Dinov2, SigLIP, DaViT, etc.
Language Encoder/Text Encoders	LLaMa, T5, LLaVA, QWEN2, etc.
Fusion Method	FiLM, concatenation, cross-attention, etc. [3]
Action Decoder/Action Head	Autoregressive, flow-matched, diffusion policy head, etc.
Embodiment	Single-arm, bimanual, quadruped, dexterous hand, humanoid, unmanned aerial vehicle (UAV), self-driving car, etc.
Training Data	DROID, LIBERO, Open-X-Embodiment, BridgeData, etc.
Training Strategy	From scratch, fine-tuning, co-fine-tuning, parameter-efficient fine-tuning, etc.
Task Approach	Low-level control policy, high-level task planner, hierarchical
Modality	Text, vision, speech, etc.
Cameras	Single camera, multiple cameras, depth camera
2D Inputs vs. 3D Inputs	2D scene image as input vs. 3D point cloud, etc.
VLM/LLM	Llama2, Fuyu, PaliGemma, LLaVA, Qwen, Prismatic, etc.
One Step vs. Historical	Current image/images or history of past images at input, etc.
Action Type	Subtasks, action primitives, low-level end-effector poses [54]

Fusion is the process of combining features from different modalities, and common approaches include feature-wise linear modulation (FiLM), concatenation, and cross-attention [3]. FiLM [55] is used in neural networks to modify intermediate feature maps

conditioned on an external input, and it is used in VLMs to fuse vision and language inputs. In essence, FiLM makes vision models respond to linguistic inputs, thus making them capable of performing adaptive reasoning. Cross-modality attention establishes a relationship between two modalities by extracting information from one modality to attend to related information in another modality. In concatenation, embeddings from different modalities are simply concatenated together.

Action decoders or action heads are mainly responsible for generating low-level robotic actions. They can be implemented using autoregressive transformers, where discrete action tokens are generated at the output one token at a time, similarly to the way text tokens are generated or continuous actions. Diffusion transformer (DiT) [56] was the initial transformer architecture for diffusion models. Diffusion based action heads as well as flow-matching action heads, generate continuous actions. Other variations could be based on the embodiments the VLA supports, what kind of training data is used, how training is undertaken, the pre-trained LLM/VLM model used, differences in supported modalities, and whether VLA is used for converting a robotic task to subtasks as a high-level planner or for generating low-level robotic actions or a combination. Certain VLAs only support a single-arm robot setup, while others can support bimanual, quadruped, and even self-driving cars as embodiments.

Some VLAs only use 2D images while others incorporate 3D information for better spatial understanding. VLAs can be trained using only the current image or multiple images, as explored in [57].

Due to the breadth and heterogeneity of the VLA literature, the studies covered in this review are organized into four tables according to their primary contribution. Table 5A summarizes the core VLA architectures and representative models that define the present research landscape. Table 5B presents studies focused on fine-tuning, architecture adaptation, and reinforcement-learning-based enhancement of VLA models. Table 5C compiles methods aimed at improving inference efficiency, compression, and deployability. Finally, Table 5D highlights the diversity in terms of modalities and embodiments supported by the VLAs included in this review.

Table 5. (**A**) Summary of selected VLA models. (**B**) VLA architecture adaptation and reinforcement-learning-based methods. (**C**) VLA inference optimization, compression, and deployment efficiency methods. (**D**) Diversity in VLA applications, modalities, and embodiments.

Reference	Model	Summary
		(**A**)
[14]	RT-1	Instructions are encoded using a universal sentence encoder, and 6 images are fed into EfficientNet-B3, followed by FiLM-based fusion to generate vision–language tokens that are compressed via TokenLearner and fed into a decoder-only transformer to generate discrete actions.
[48]	RT-2-PaLI-X-55B/ RT-2-PaLM-E-12B	PaLI-X and PaLM-E VLMs co-fine-tuned using web-scale visual question answering data and robotic action data have resulted in VLAs that exhibit good generalization, reasoning, and emerging capabilities.
[51]	OpenVLA	An open-source VLA developed using LLaMA 2 as a text encoder and SigLIP and DinoV2 as vision encoders. Quantization and parameter-efficient fine-tuning were explored to enable training and inference on consumer-grade GPUs.
[58]	LEO	A generalist multimodal agent that takes text, 2D images, and 3D point clouds as input and can undertake robotic manipulation in a 3D environment, as well as embodied navigation.
[59]	3D-VLA	VLA is trained on a large-scale 3D embodied instruction set that uses a pre-trained BLIP2-FlanT5 model, a novel set of interaction tokens, and diffusion models for point cloud and goal image generation.

Table 5. *Cont.*

Reference	Model	Summary
[60]	Bi-VLA	A VLA for bimanual robotic manipulation that uses a pre-trained Starling-7B LLM for semantic planning, code generation, and motion control and QwenVL to identify if an object is present in the scene and, if it is present, what the 2D coordinates are. No training or fine-tuning is undertaken in this approach.
[61]	SC-MLLM	A CLIP vision encoder and a projection layer are used to transform images into embeddings of a 7B LLaMA LLM to predict end-effector poses, which are then used by a failure detection and correction module, and the model leverages continual policy learning.
[62]	RoboMamba	RoboMamba combines a CLIP vision encoder and a selective state space-based Mamba LLM in an efficient VLA architecture that has good reasoning and manipulation capabilities and faster inference speeds.
[63]	A3VLM	An object-centric articulation-aware model that does not require robotic interaction for training. Instead of low-level actions, A3VLM identifies 3D bounding boxes, joint types, joint axes, and action types, thus making it more generalizable for different robots. Use CLIP, DinoV2, QFormer, and LLaMA.
[64]	HumanVLA	A VLA for humanoid robots to arrange objects in a room-like environment. VLA is trained using a teacher model derived through reinforcement learning and adversarial motion prior, which is distilled into VLA using behavioral cloning.
[65]	LLaRA	Created a visuomotor instruction tuning dataset from a behavior cloning dataset and fine-tuned LLaVA on this dataset. Another difference is that the action decoder generates the position of the object in image coordinates and a rotation component.
[66]	Actra	Introduced trajectory attention with learnable action queries inserted between the state and action segments. Used a contrastive learning objective to improve multimodal encoding.
[67]	HiRT	Hierarchical transformer model that uses a System 1- and System 2-based approach, where slow high-level feature extraction from VLM is decoupled from fast low-level conditioned action generation policies.
[68]	N/A	A novel approach where a VLA is combined with a diffusion policy for manipulation using a 13 DoF robotic hand.
[69]	DP-VLA	A dual-process, System 1 and System 2 approach, where System 2 is a pre-trained VLM or VLA that generates latent features using lingual instructions and images as inputs. The latent features are used by a System 1 component, such as BC-Transformer, for high-frequency task execution.
[70]	π_0	A robotic foundation model pre-trained on large and diverse cross-embodiment data, followed by task-specific fine-tuning, can undertake very complex manipulation tasks such as shirt folding, grocery bagging, etc. A flow-matching-based action expert is used to generate continuous actions.
[71]	N/A	A VLA model in which the reasoning capabilities of the language model are leveraged to find the target object in the scene and end-effector motion before generating grasping actions.
[72]	CLIP-RT	Uses language supervision and contrastive imitation learning to train a VLA. A framework to collect robotic data through language supervision and a stochastic trajectory augmentation process was also proposed.
[73]	CogAct	A robotic foundation model that uses learnable cognition tokens as part of the VLA architecture and conditions action on these tokens through a diffusion transformer-based action model. At inference, an adaptive ensemble strategy is employed where the action at the current time also uses weighted historical predictions for the current timestamp.
[74]	Diffusion-VLA	Combines autoregressive next-token prediction for reasoning with a diffusion model for action sequence generation to train a visuomotor policy.
[75]	ObjectVLA	Employed a diffusion-based approach with additional bounding box data when co-fine-tuning with robot data.

Table 5. *Cont.*

Reference	Model	Summary
[76]	DexGraspVLA	A hierarchical model that uses a high-level task planner to generate affordance signals that are used by the low-level controller to generate dexterous grasping actions through a diffusion-based head. Uses DINOv2 to extract domain-invariant features.
[77]	OTTER	Extracts text-aware visual features using pre-trained CLIP.
[78]	PointVLA	3D features are injected into pre-trained VLAs to improve their performance.
[79]	MoRE	Mixture of robotic experts (MoRE) uses a mixture of LoRA experts fine-tuned for different tasks. MoRE is trained using reinforcement learning and has shown good results on hard tasks for quadruped robots.
[80]	MoManipVLA	A framework to transfer pre-trained VLA policies to mobile manipulation tasks.
[81]	HybridVLA	Combines autoregressive and diffusion denoising action generation.
[82]	Dita	Dita is a diffusion transformer policy that denoises continuous action sequences via in-context conditioning.
[83]	Gemini Robotics	Uses embodied reasoning with cloud-based and onboard components. Can be easily adapted to new tasks and embodiments.
[84]	OPAL	OPAL uses an innovative topological attention mechanism to enforce physical constraints in addition to fourth-order Runge–Kutta integration to generate a long-horizon action sequence with good inference speed.
[85]	π0.5	A VLA recipe that uses hierarchical high-level subtask prediction and low-level continuous action generation using flow matching. The training recipe consists of pre-training on diverse multimodal web and robot data and post-training for mobile manipulation.
[86]	GraspVLA	A foundational model for grasping tasks trained on large-scale synthetic data.
[87]	VLA-Grasp	A task-oriented grasping model that uses point cloud, text, and action data to generate grasping poses that are used to generate actions.
[88]	UniVLA	Uses a task-centric latent action space to create an embodiment-agnostic generalist policy. The approach significantly reduces pre-training compute and data requirements.
[89]	OE-VLA	Open-Ended VLA (OE-VLA) allows multimodal instructions combining text and images instead of just text-based inputs to VLA.
[90]	OneTwoVLA	A unified model that has System 1 (acting) and System 2 (reasoning) capabilities. The key strengths are long-horizon task planning and error detection and recovery.
[91]	Hume	Modeled on value-guided System 2 thinking combined with a lightweight System 1 to perform actions, Hume has shown promising average success rates at high execution frequencies.
[92]	ChatVLA-2	It uses a mixture-of-experts (MoE) module along with a pre-trained VLM with some task-specific experts working with experts that share common multimodal features.
[93]	Agentic Robot	A framework that uses high-level task planning through a large reasoning model, VLA for task execution, and a VLM based on verified information to decide whether to continue or retry an action.
[94]	LoHoVLA	A unified VLA framework that generates language tokens for subtask generation in addition to the action tokens.
[95]	OG-VLA	OG-VLA combines multiple RGBD observations projected into orthographic views that are processed by a vision backbone and an LLM and then decodes action tokens via an image generator component to create heatmap annotations, which are decoded to obtain the actions.
[96]	SwitchVLA	A framework to train instruction-conditioned policies that allow switching behavior using contact state and behavior mode as additional supervisory signals.
[97]	GF-VLA	Graph-Fused VLA combines a scene graph extracted from demonstration data with VLA reasoning to deliver strong robustness and generalization capabilities.

Table 5. *Cont.*

Reference	Model	Summary
[98]	RationalVLA	This is a dual system model that combines a high-level visual language model with low-level manipulation policies to enable robots to reject a user instruction if the instruction is defective or can have safety implications.
[99]	ControlVLA	A few-shot learning approach that uses a pre-trained VLA with object-centric representation, thus giving a good success rate with only 10–20 demonstrations.
[100]	UniVLA	Facilitates unified vision–language–action learning in which all modalities are transformed into discrete tokens while sharing a common vocabulary.
[101]	4D-VLA	4D-VLA fuses spatiotemporally aware vision tokens with task instructions for action prediction.
[102]	cVLA	A simple model to predict trajectory in image-frame coordinates instead of end-effector action prediction, making it embodiment-agnostic.
[103]	TriVLA	Augments the traditional dual system models (System 1 and System 2) with a third system that acts as the world model for current and future prediction states.
[104]	DreamVLA	DreamVLA undertakes world knowledge forecasting by predicting semantic knowledge, depth map, and dynamic regions that improve action planning.
[105]	EgoVLA	Uses egocentric videos of humans performing a task during training to convert into human action. The human actions are converted into robot actions by converting the human wrist location into the robot end-effector location.
[106]	InstructVLA	Uses a mixture-of-experts (MoE) LoRA adaptation with a pre-trained VLM to switch between latent action generation and multimodal reasoning. Latent actions are fed into the flow-matching action expert to generate robot actions.
[107]	GR00T N1	An open foundation generalist model for humanoid robots; GR00T uses a dual system approach, with System 2 interpreting the environment and System 1 generating actions.
[108]	SpatialVLA	Injects spatial information by converting 2D images to 3D position encodings and converting continuous robot actions to discrete actions. This combination improves embodiment generalization.
[109]	ChatVLA	Uses the mixture-of-experts (MoE) approach coupled with phased training, where in the first stage, the model is only trained using robot manipulation data, followed by co-training on a mixture of robot and text–image paired data.
[110]	FLARE	A framework that combines latent world modeling with robot policy learning to improve long-term reasoning of VLAs
[111]	Evo-0	A framework that enhances spatial understanding of VLAs by adding a plug-and-play module to add 3D geometry features via a visual geometry foundation model instead of requiring a depth camera or point cloud data.
		(**B**)
[112]	TraceVLA	Fine-tuned OpenVLA using visual traces of historical trajectories in addition to the current camera observation to improve the spatiotemporal awareness of the model.
[113]	Emma-X	Fine-tuned OpenVLA using hierarchical embodiment data that consists of 3D spatial movement needed to obtain the end state, 2D gripper position, and grounded task reasoning.
[114]	CoA-VLA	Using DiffusionVLA as a base, reasoning is injected that improves visual affordance. In addition to accuracy improvements, this also improves collision avoidance for robotic manipulation tasks.
[115]	iRe-VLA	At the fine-tuning stage, training is carried out by iterating between online reinforcement learning and reinforcement learning.
[29]	ConRFT	A fine-tuning method that uses both offline and online reinforcement learning with human feedback to improve VLA performance for specific tasks.
[116]	Expert-VLA	Uses a human in the loop as an expert who assists the VLA policy, thus generating datasets that can be used in further fine-tuning of the VLA.
[117]	RPD	Refined policy distillation extracts compact task-specific expert policies from generalist VLA policies.

Table 5. *Cont.*

Reference	Model	Summary
[118]	ReBot	A real-to-sim-to-real pipeline that scales robot data to improve VLA accuracy.
[119]	CoT-VLA	Chain-of-thought VLA uses a visual chain-of-thought reasoning to predict sub-goal future images and then generates actions to achieve those goals.
[120]	ASK-To-ACT	A framework to train an agent to ask questions in response to human instruction for disambiguation. Reinforcement learning with rewards generated by an LLM is used for training the agent.
[121]	TOPIC	Task-prompt graph evolution policy known as TOPIC is a few-shot continual learning framework that uses task-specific prompts and a continuous evolution strategy to build a task relation graph. It helps avoid catastrophic forgetting.
[122]	Interleave-VLA	A method to allow existing VLAs to process interleaved image–text tokens that improves generalization capability but requires more training steps to converge.
[123]	3D-CAVLA	Uses chain-of-thought reasoning, the PointNet depth encoder, and region-of-interest pooling to improve zero-shot capabilities.
[124]	ReinboT	VLA that uses the reinforcement learning principle of maximizing cumulative reward.
[125]	RoboFAC	A model that can be integrated with VLAs for failure understanding, analysis, and correction.
[126]	InSpire	Intrinsic spatial reasoning, known as InSpire, is a method devised to enhance the spatial reasoning ability of autoregressive VLAs by appending an additional object-direction-related question with textual task instruction tokens.
[127]	RIPT-VLA	Uses post-training reinforcement learning with sparse binary rewards to improve VLA performance in low-data settings.
[128]	VLA-RL	A framework to improve the performance of autoregressive VLAs using online reinforcement learning.
[129]	ReFineVLA	A large teacher VLM provides visual cues, task planning, and spatial reasoning to VLA during fine-tuning to improve performance.
[130]	RFTF	RFTF is a reinforcement-learning-based approach to fine-tune VLAs but uses dense reward during value function learning, which leads to improved generalization.
[131]	KitchenVLA	Improves robotic task execution in kitchen environments by taking human instructional videos and using a VLM to identify the gap between the robot and video domains. These mismatches are passed on to an LLM-based planner that suggests corrective steps.
[132]	TGRPO	Introduced a new online reinforcement learning algorithm called trajectory-wise group relative policy optimization for fine-tuning a pre-trained VLA.
[133]	CO-RFT	A two-stage policy training method, with the first stage using behavior cloning with full parameter fine-tuning, followed by offline reinforcement learning with action chunking.
[134]	RICL-VLA	Retraining for in-context learning (RICL) is a system that uses RAG and a small robot demonstration dataset to empower an existing VLA with in-context learning capabilities.
[135]	ReVLA	Used model merging to revert weights of visual encoders of robotic foundation models to their original VLM version to mitigate catastrophic forgetting and improve visual generalization.
[136]	DyCRIS	A grasp pose estimation model that uses RGB image, depth image, and language instruction as inputs to a pipeline consisting of multiple models: CRIS for image segmentation, CLIP for language-guided object localization, Mask R-CNN for segmentation, 3D-GCN for point cloud refinement, and SAR-Net for pose estimation.
		(**C**)
[137]	TinyVLA	A fast and data-efficient VLA that uses LoRA and a diffusion policy head with a smaller VLM.
[138]	DeeR-VLA	A novel approach where the VLA architecture allows early exit at inference time based on a termination condition, thus saving compute and memory requirements.

Table 5. *Cont.*

Reference	Model	Summary
[139]	QUART-Online	A low-inference-latency model for quadruped robots that uses action chunk discretization.
[140]	VLA-cache	A strategy where VLA inference is improved by caching the static tokens that are similar between input images at different timesteps and selecting the dynamic tokens, thereby increasing the inference speed by 1.7 times.
[141]	OpenVLA-OFT	Combined action chunking and parallel decoding strategies to increase the inference speed.
[142]	FLOWER	FLOWER uses a compact VLM combined with a flow transformer, thus requiring less GPU memory and compute. This approach also reduces the pre-training time.
[143]	PD-VLA	Action chunking and parallel decoding are used to increase the speed.
[144]	MoLe-VLA	Mixture-of-layers VLA employs a spatiotemporally aware router for layer skipping and cognitive self-knowledge distillation to enable a computationally efficient VLA.
[145]	NORA	NORA is an efficient model that can be fine-tuned on a consumer-grade GPU and employs the FAST+ tokenizer.
[146]	RT-cache	This is a pipeline that builds a memory of a large number of embedded image–action pairs in a trajectory database and searches the current frame of the camera at runtime to find the most similar trajectory and play it, thus reducing compute requirements at the inference stage.
[147]	FLASHVLA	An inference speed acceleration framework that uses similarity in consecutive actions and visual token redundancy to reduce compute requirements at the inference stage.
[148]	EaqVLA	EaqVLA is a framework that uses a mixed-precision quantization strategy with 4-bit quantization applied to vision modules and 8-bit quantization applied to the language module.
[149]	SQIL	Saliency-aware quantization for imitation learning that gives better performance than post-training quantization and quantization-aware training.
[150]	SmolVLA	A compact and efficient VLA that discards the last layers of VLM and uses interleaved cross-attention- and self-attention-block-based action experts and asynchronous inference for running VLA on low-cost robots.
[151]	Fast ECoT	This is an inference acceleration method that caches and reuses high-level reasoning and parallelizes the reasoning to decrease latency.
[152]	BitVLA	BitVLA is a 1-bit VLA model that uses distillation-aware training to quantize the vision encoder to 1.58 bits and reduces memory usage, making it suitable for deployment on memory-constrained edge devices.
[153]	SP-VLA	A dual system where the VLA model generates deliberative actions while intuitive actions are delegated to a lightweight model. This approach, in addition to token pruning, leads to inference speed improvements.
[154]	RLRC	A framework that uses structured pruning, 4-bit quantization, and performance recovery via supervised fine-tuning and reinforcement learning. It reduces the memory requirements by 8× and increases the inference speed by 2.3×.
[54]	VQ-VLA	A method that leverages a vector-quantization-based action tokenizer to improve inference speed.
[155]	VOTE	An inference optimization technique and a training framework that generates a single special token instead of multiple tokens. Action is selected through the voting of the action committee.
[156]	BYOVLA	A runtime intervention scheme that finds and edits task-irrelevant regions in the input to improve the success rate of VLAs when there are visual distractors in the scene.
[157]	EdgeVLA	This approach uses small language models combined with predicting the entire end-effector position at once instead of autoregressive prediction, thus making this recipe suitable for real-time performance on edge devices.

Table 5. *Cont.*

Reference	Model	Summary
		(**D**)
[158]	RoboNurse-VLA	A novel VLA that allows robots to act as scrub nurses and hand over surgical instruments to doctors in response to voice-based instructions. The pipeline uses Gladia for speech recognition, YOLOv8 for bounding boxes of instruments, SAM-2 to create image segments, and MLP to project visual tokens to the LLaMA 7B embedding space to predict the next action token.
[159]	VLAS	First VLA to incorporate speech in the model itself without relying on external speech recognition, thus allowing speech-based communication with robots.
[160]	Shake-VLA	A VLA that uses speech and force torque sensors with a bimanual robotic setup to prepare cocktails.
[161]	Humanoid-VLA	A framework to enable humanoid robot control that uses egocentric vision and enhanced contextual understanding to facilitate adaptive object interaction.
[162]	Safe-VLA	Introduced an integrated safety approach that leverages constrained Markov decision process (CMDP)-based reinforcement learning to ensure that the policy takes into account safety requirements.
[163]	OpenDriveVLA	VLA for end-to-end autonomous driving with spatial reasoning and trajectory planning capabilities.
[164]	EndoVLA	EndoVLA combines supervised fine-tuning and reinforcement fine-tuning with verifiable rewards to fuse endoscopic vision and surgeon prompts for autonomous endoscopic tracking through continuum robots.
[165]	DriveMoE	A mixture-of-experts approach for end-to-end autonomous driving that dynamically selects relevant camera inputs and only activates the expert needed for specialized planning.
[166]	ForceVLA	This is a framework that integrates an additional modality of 6-axis force feedback along with vision, language, and proprioception to predict action trajectory.
[167]	BridgeVLA	BridgeVLA converts 3D inputs to 2D images to align them with the input requirements of a pre-trained VLM, which generates a 2D heatmap in the pre-training stage. In the fine-tuning stage, the model takes three orthographic projections of the 3D point cloud and generates heatmaps to show the position of the end effector in the next frame.
[168]	Tactile-VLA	Treats tactile sensing as an additional modality to enable precise force control when generating actions.
[169]	VLA-Touch	Enhanced VLA models through tactile-assisted task planning, as well as tactile-assisted policy refinement for manipulation.
[170]	FedVLA	A federated learning framework for VLAs to preserve the privacy of the data.
[171]	QUAR-VLA	A VLA designed for quadruped robots for spatial navigation, obstacle avoidance, environment adaptation, and object manipulation. Co-fine-tuned Fuyuna-8B on the QUARD dataset.
[172]	UAV-VLA	UAV-VLA uses satellite images and VLM and LLM to generate flight paths for unmanned aerial vehicles (UAVs) based on text-based user requests.

Some of the models included in this study laid the foundations for VLA research, such as RT-1, RT-2, and OpenVLA, and therefore, they have been cited more than the others. OpenVLA, being the first open-source VLA, also became the most common model used in comparisons with subsequent research. A graph of citations is given in Figure 5, showing the relationship between different studies. Most frequently cited studies are shown in dense color with lines and arrows showing relationship between citing and cited papers.

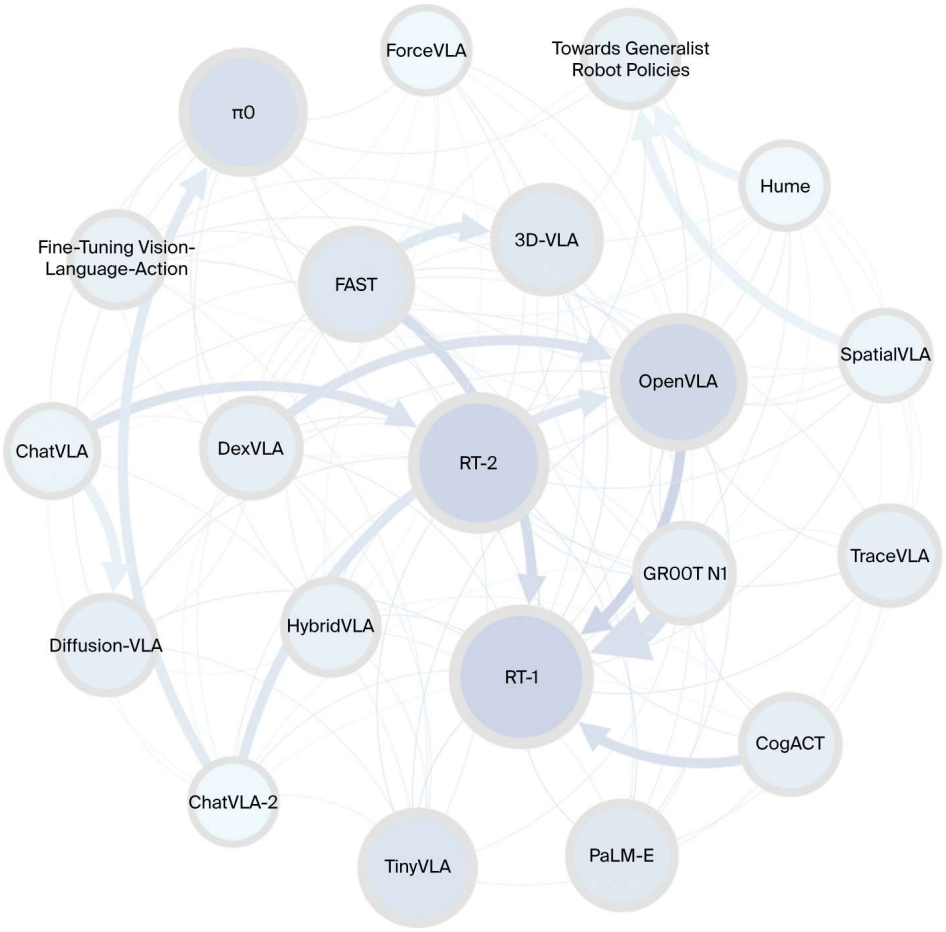


Figure 5. Citation graph showing the relationship between models covered in the most cited studies.

4.2. VLA Applications

VLA applications can be broadly classified into high-level and low-level applications. At the high level, VLAs serve as generalist robotic policies across a range of domains, including autonomous vehicles, healthcare and medical robotics, industrial automation, humanoid robotics, and precision agriculture [9]. They are also being explored for service and hospitality applications. For example, Shake-VLA [160] enables a robotic system to prepare cocktails, whereas KitchenVLA [131] supports the execution of various tasks in kitchen environments. VLAs also show potential for household robotics. The ASK-TO-ACT framework presented in [120], for instance, employs a VLA-based embodied agent that performs household tasks while asking clarifying questions when additional information is required. DriveMoE [165] is an example of a VLA for end-to-end autonomous driving, and UAV-VLA [172] is used to generate flight plans for unmanned aerial vehicles [UAVs]. RoboNurse-VLA [158] is a model for applications in the medical domain, where it acts as a scrub nurse and hands over surgical instruments to doctors based on speech-based instructions. HumanVLA [64] and Humanoid-VLA [161] are humanoid robots that explore their environment and interact with objects.

Low-level applications of VLAs include robotic manipulation, language-conditioned grasping, task and motion planning, autonomous pick and place, multimodal reasoning, and complex instruction grounding.

4.3. VLA Training Recipes

VLA training is a resource-intensive task [173]; however, VLA training recipes differ in terms of training approach, training datasets, loss function, size and kind of pre-trained

model, and whether any optimization is used during training. There are three main training approaches, namely training from scratch, fine-tuning, and co-fine-tuning. In a training-from-scratch approach such as the one used in RT-1 [14], only robotic demonstration datasets are used to train the model. In the fine-tuning approach, a pre-trained VLM is fine-tuned using robotic demonstration data. An example of this approach is OpenVLA [51], where the Prismatic-7B VLM was fine-tuned using Open X-Embodiment robotic data. In the co-fine-tuning approach, the model is trained on web-scale vision–language training data and robot action data. RT-2 [48] is an example of a VLA trained using co-fine-tuning where both web-scale datasets and robot datasets are used. Often, VLAs are trained for a single embodiment, but if the training dataset has multi-embodied robot demonstrations, then such training is called cross-embodied training, as is carried out in π_0 [70], DexVLA [174], and UniVLA [88].

Fine-tuning can also be categorized into full-fine-tuning, where all weights are updated, and parameter-efficient fine-tuning (PEFT) methods [175], where trainable parameters are reduced, thus reducing the computational requirements to train the model. Some training paradigms also include reinforcement learning-based fine-tuning to improve the performance of VLAs such as ConRFT [29].

In terms of training datasets, the datasets can be web-scale image–text data; openly available robotic demonstrations data, such as Open-X-embodiment [49] data; or demonstrations collected using teleoperation. Training data can also be sourced from video demonstrations. Training data is often classified as simulated or synthetic data and real data. There are many environments that generate simulated data, such as Issac Gym and Isaac Lab [176]. The most common datasets used are given in Figure 6. Although most datasets focus on manipulation tasks, there are examples of datasets that are very specialized, such as ScanBot [177], which consists of trajectories for laser scanning to train VLAs to undertake precise laser scanning. Another unique dataset combination is employed by Wen et al. [178], where 3D robot state estimation is combined with expert demonstrations.

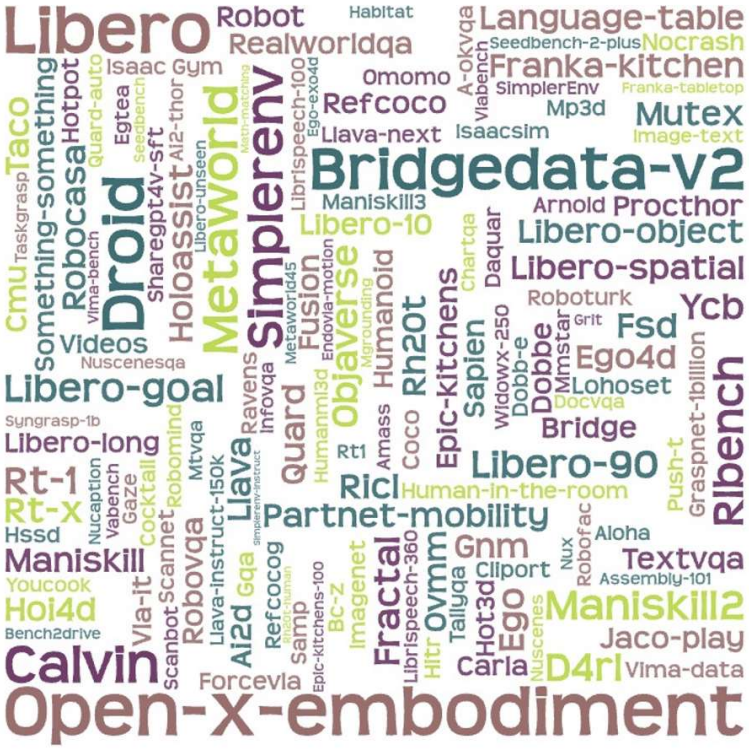


Figure 6. Datasets commonly used in VLA training.

Another variation is in terms of which VLM is fine-tuned, and often, this plays a major role in defining the size of the VLA model in terms of the number of parameters. Examples of small-sized VLAs include TinyVLA [137] and SmolVLA [150], both with fewer than 1.5 billion parameters, whereas examples of large-sized VLAs include RT-2-PaLI-X-55B, with 55 billion parameters. A comparison of selected models in terms of the number of parameters is given in Table 6.

Table 6. Comparison of selected models in terms of size.

Model Name	Number of Parameters (Million/Billion)	Model Name	Number of Parameters (Million/Billion)	Model Name	Number of Parameters (Million/Billion)
RT-1	35 M	Emma-X	7 B	Hume	<5 B
RT-2-PaLI-X-55B	55 B	QUART-Online	8 B	SmolVLA	0.45 B
RT-2-PaLM-E-12B	12 B	ObjectVLA	2 B	GR00T N1	2 B
LEO	7 B	FLOWER	950 M	π_0	3.3 B
RoboMamba	3.2 B	HybridVLA	2.7 B 7 B	CLIP-RT	1 B
OpenVLA	7 B	Dita	334 M	CogAct	7 B + 330 M
Actra (Large)	Large 198 M	CoT-VLA	7 B	DiVLA (Large)	7 B
Actra (Base)	43.3 M	OpenDriveVLA (Large)	7 B	DIVLA (Small)	2 B
Actra (Small)	19.4 M	OpenDriveVLA (Base)	3 B	NORA	3 B
TinyVLA (Huge)	1.3 B	OpenDriveVLA (Small)	0.5 B	EdgeVLA	1 B
TinyVLA (Base)	740 M	UniVLA	7 B		
TinyVLA (Small)	422 M	RoboFAC	7 B		

Most VLAs support vision and language as input modalities, but some VLAs, such as FuSE [179], support additional input modalities, such as touch and audio. Intelligent integration of force feedback as a modality in VLAs can also result in more robust robotic manipulation [166]. Speech encoders can also be employed in VLA architecture to train VLAs to undertake actions in response to human speech.

Voice-based retrieval augmented generation (RAG) enables the same embodiment with the same speech to act differently based on differences in the voiceprint [159]. Tactile sensing can also be used as an input modality, in addition to vision and language [168].

4.4. VLA Evaluation and Benchmarking

VLAs are evaluated using simulated benchmarks as well as on real robots. Some of the key metrics when evaluating the performance of VLAs are Success Rate, Inference Speed, Control Frequency, Inference Latency, Time to complete task, Plan Success Rate, Memory usage, and Computational cost. While evaluating VLA performance, they are often tested for generalization performance, zero-shot performance, and robustness to distractors and changes in background conditions. Table 7 shows the main metrics used for VLA evaluations. For evaluating VLA performance on different tasks, several benchmarks have been proposed. The most frequently used benchmarks include LIBERO, SimplerEnv, and CALVIN. Figure 7 shows the benchmarks used for evaluating VLAs. LIBERO [180] stands for Lifelong Learning Benchmark on Robotic Manipulation, and it has four task suites, namely LIBERO-SPATIAL, LIBERO-OBJECT, LIBERO-GOAL, and LIBERO-100. SimplerEnv [181], also known as the SIMPLER benchmark, includes a set of simulated

environments for the evaluation of robotic manipulation tasks. CALVIN [182] is another simulated environment for evaluating language-conditioned long-horizon tasks.

Table 7. VLA evaluation metrics.

Evaluation Metric	Description
Success Rate	A metric that measures if the instruction is performed successfully by the robot. Often, it is averaged over multiple tasks. At times, it is reported for both seen tasks and unseen tasks to highlight the generalization performance of the VLA. Higher values are better.
Inference Speed	It measures the number of inferences per second, such as action tokens per second. A higher value is better.
Control Frequency	The rate at which VLAs generate output commands for robotic control. It is measured in Hertz (Hz). Higher values are better.
Inference Latency	Inference latency refers to the time taken to produce an output after receiving a multimodal input. It is measured in milliseconds or seconds, and lower values are better.
Time to Complete Task	Time taken to complete a task assigned to the robot. Measured typically in seconds, and lower values are better.
Plan Success Rate	When a model generates a high-level plan, it evaluates if the skills selected by the model are what was intended by the instruction [21].
Memory Usage	This refers to the amount of memory consumed by the model during the inference stage.
Computational Cost	This is mostly measured as floating operations per second (FLOPS); lower values are better.
Generalization	Generalization measures the effectiveness of a model in handling inputs it has not seen during training.
Robustness	This typically measures the effectiveness of the model in the presence of distractors in the scene, change in the background, change in the camera position or when illumination conditions change in the environment.
Mean Squared Error	Mean Squared Error is used to compare the difference between the ground truth and observed trajectory of the robot.

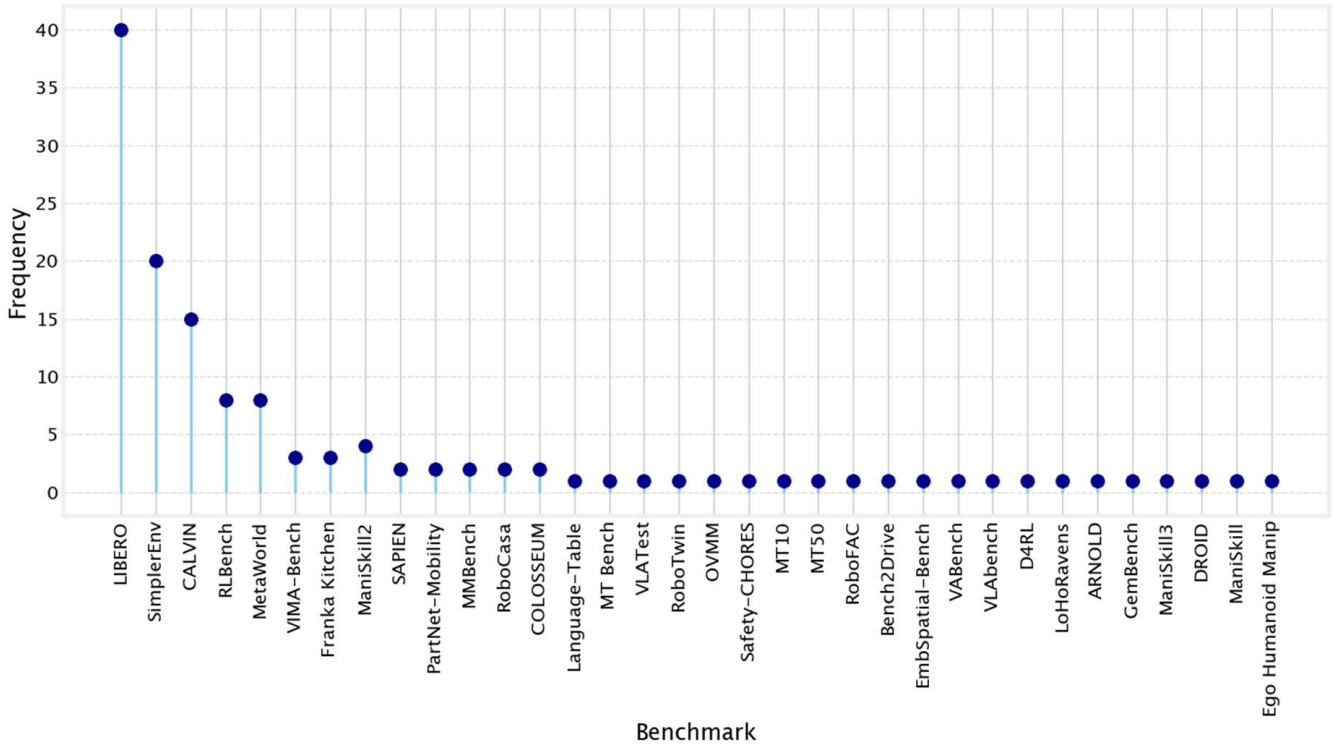


Figure 7. Commonly used benchmarks in VLA research.

In addition to simulated environments, most VLAs are also tested on real robots. There is a wide range of embodiments on which VLAs have been tested, including robotic arms, bimanual setups, quadrupeds, mobile robots, and even humanoids. The most widely used setup for evaluation is the robotic arm, whereas quadruped and humanoid embodiments are less common in VLA research at the time of this analysis. Figure 8 shows the distribution of embodiments and the distribution of robotic arms used in VLA research.

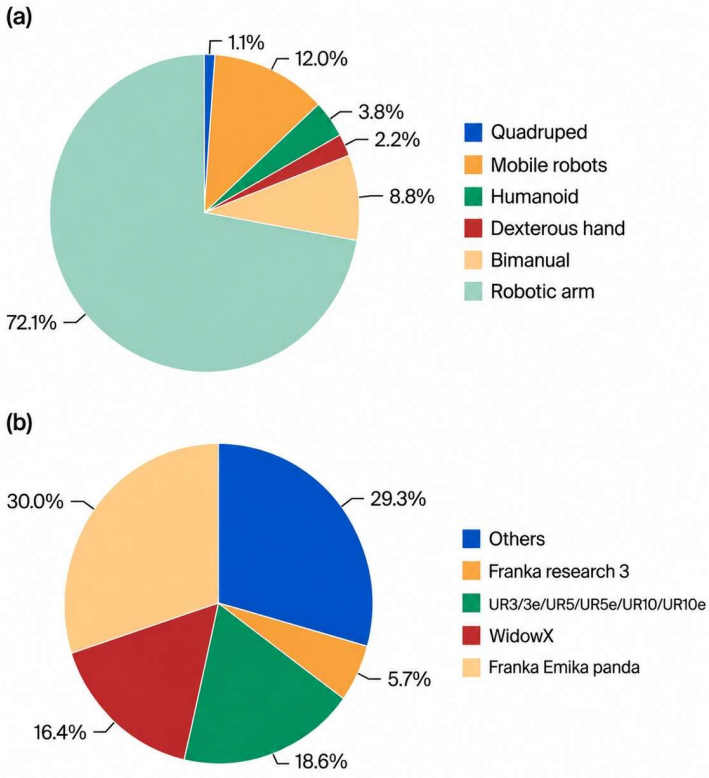


Figure 8. Robots and embodiments: (**a**) distribution of embodiments and (**b**) common robots used in VLA Research.

A list of robots mentioned in the literature covered in this review is given in Appendix B. Most studies were focused on single-arm or bimanual robotic setups, with very few studies carried out on deploying VLAs in quadruped robots and humanoids. There is also not much coverage of deploying VLAs on low-cost hardware, except for SO-100 and SO-101 robotic arms, which were only used in SmolVLA [150].

4.5. Best Practices for VLA Development

The studies included in this review have identified some common techniques to improve the performance, robustness, and generalization of VLA models. The following are some of the key findings.

It has been observed that data diversity is more important than data quantity for improving the generalization of robotic policies [14].

Co-fine-tuning, where the model is trained on web-scale vision–language training data and robot action data, results in better generalization than pure fine-tuning, where the model is only trained on robot action data [48]. Similarly, An et al. [62] discovered that co-training VLA with robotic data and general instruction data enhances the general scene reasoning capability of VLAs. Spatial relationship based visual chain-of-thought methods also improve generalization significantly [183]. As per Li et al. [184], vision-language pre-training is a key factor enabling strong VLA performance. They also found that large-scale cross-embodiment pre-training followed by post-training on in-domain data yields

better performance. Another way to improve performance on specific tasks is to undertake task-specific fine-tuning of a VLA using reinforcement learning [29].

In terms of the vision encoder, it was observed in OpenVLA [51] that, when a pre-trained DinoV2 is used with SigLIP for visual encoding, it enhances the spatial reasoning of the VLA. They also found that fine-tuning the vision encoder instead of the frozen vision encoder was better for VLA performance. Some VLAs use RGB-D images, which provide depth information. This 3D information from depth improves the spatial understanding of VLAs and renders them robust to variations in spatial layouts, simultaneously improving the robustness of the model for unseen tasks [108,123]. Improving spatial reasoning capabilities can also improve generalization performance [126]. Another effective recipe for using this additional spatial information is to convert a point cloud derived from RGB-D into grasp poses using a pre-trained model based on GraspNet-1B in order to potentially improve grasping performance.

The reasoning ability of VLA is also an important factor that can improve performance. Injecting reasoning in VLAs using the chain-of-affordances mechanism improves generalization and collision avoidance [114], whereas embodied chain-of-thought reasoning improves the generalization of the policy [151,185].

VLAs that use historical (visual and proprioceptive) context and a continuous action space are better in performance than one-step VLAs with a discrete action space. The benefits of adding action history when learning an action policy were also mentioned in [65].

When it comes to the action decoder, the autoregressive generation of action tokens is compute-intensive, and using a diffusion policy action head instead of discrete action tokens is more data-efficient [137]. Li et al. [73], however, found that replacing the action head with large diffusion transformer-based action modules and action ensembling may improve performance.

Action chunking and parallel decoding can also improve inference speeds of VLAs [143]. For dexterous robotic actions at high control frequency using VLAs, action chunking with flow matching has also been used effectively [70]. Action chunk discretization has been found to improve the inference latency [139]. Lastly, the training speed of VLAs can be improved using action token compression techniques [186]. Li et al. [187] used relative action chunks to improve VLA generalization.

Several training-free and flexible modular frameworks have also been proposed to enhance existing VLAs and improve the performance of generalist robotic policies. For example, PointVLA [78] enables the integration of 3D world knowledge into existing VLAs by incorporating point-cloud information. VLA-Cache [140] accelerates inference by reusing cached static visual tokens across consecutive frames. BYOVLA [156] improves robustness by identifying and modifying task-irrelevant regions in visual observations. RoboVLMs [184] facilitates the integration and systematic comparison of different VLM backbones within robotic policy frameworks. Similarly, Scalable Visual Language Robotics (SVLR) [188] provides a modular framework that supports the integration of different VLMs, LLMs, and segmentation models, thereby offering an alternative to monolithic VLA architectures.

5. VLA Optimization

VLMs used to develop VLAs can have hundreds of billions of parameters, such as RT-2, where the largest model had 55B parameters and could not be deployed on an on-robot GPU; instead, they were deployed in the cloud, and the output was received over the network [48]. Model size can also affect the latency requirement of running models on a real robot [3].

RT-1 [14] increased inference speeds using a TokenLearner that reduces the number of tokens generated by EfficientNet, and it computes the tokens once and reuses them for future inferences. Since then, several techniques have attempted to reduce the model's size and/or decrease inference latencies without significantly impacting the generalization performance of VLAs.

5.1. Quantization

Most VLAs use transformer architecture under the hood with billions of parameters stored in 32-bit floating-point (FP32) format. Quantization is a technique in which weights and activations of the model are converted into low-precision data types, such as 8-bit integer (INT8) or 4-bit integer (INT4). This change to low-precision data types allows faster matrix multiplication while reducing the storage memory requirement. OpenVLA [51] was the first VLA that used low-rank adaptation (LoRA), a parameter-efficient fine-tuning (PEFT) method, and quantization to increase compute efficiency during training and inference stages, respectively. A 4-bit quantization scheme reduced the graphics processing unit's (GPU) memory requirement by half without any performance loss compared to the bfloat16 precision model. In [188], a Phi3-mini-4k-instruct LLM was quantized to 4-bit, and 8-bit quantization carried out by Zhang et al. [144] resulted in an inference frequency of 15.7 Hz compared to 9.8 Hz for the FP16 version. This increase in inference frequency was coupled with a reduction in GPU usage to 55% compared to that of the FP16 version.

While quantization reduces the model's size, it can result in task failure in complex robotic manipulation tasks [148]. The encoding-aligned quantization method of EaqVLA [148] uses module-specific quantization, where vision encoder and projector modules can be quantized to INT8 or INT4, but the language model is kept at higher precision for optimal performance.

5.2. Pruning

Pruning makes neural networks more sparse by removing less important weights during training, thereby allowing model compression [189]. The two most common pruning methods are magnitude pruning and movement pruning. In magnitude pruning, the weights that are farthest from zero are considered important, whereas in movement pruning, the weights that increase during fine-tuning are considered more important [183]. Pruning can be combined with quantization to further reduce the size of the model, as demonstrated in DeeR-VLA [138].

5.3. Early Exit and Dynamic Layer Activation

The intuition behind early exit is that most embodied tasks are such that they can be undertaken by smaller models, with only rare situations requiring the use of the full capacity of the model. Thus, DeeR-VLA [138] devised an MLLM architecture with intermediate exit points where computation is only carried out until the correct robot action is obtained; hence the name early exit, as forward propagation stops at a specific layer. In mixture-of-layers VLA (MoLe-VLA) [144], each LLM layer is considered an expert, and a spatiotemporally aware router is used to dynamically select the relevant layers for given input features. This approach reduced computational costs while improving model performance at the same time.

5.4. Knowledge Distillation

Knowledge distillation is another model compression technique in which knowledge is transferred from a larger, better-performing teacher model to a smaller student model during training or fine-tuning [189].

In the refined policy distillation (RPD) method, distillation is used to convert a large generalist VLA teacher into a compact RL expert policy [117]. In MoLe-VLA, knowledge is distilled from the original full-layer model to a layer-skipping student model [144]. BitVLA uses a 1-bit VLM trained using a 1-bit LLM alongside a full-precision vision transformer (ViT), followed by distillation-aware training where full-precision BF16 ViT serves as the teacher model for the 1-bit ViT [152].

5.5. Action Chunking and Compression

Autoregressive transformer-based action decoders generate one action token at a time; however, grouping robotic actions can be more effective for robotic control. This grouping of actions is called action chunking. In vector-quantized action chunking, continuous action sequences are compressed into a sequence of discrete action tokens. MiniVLA [190], a 1B parameter model with performance comparable to OpenVLA, employs vector-quantized action chunking.

Action compression has also been found to be quite effective, with action chunk discretization (ACD) used to compress actions [139]. Similarly, discrete cosine transform (DCT)-based compression has also been used for compressing robot actions to reduce the training time and learn dexterous high-frequency tasks [186].

5.6. Diffusion Head and Flow Matching

In diffusion-head-based VLAs such as TinyVLA, instead of next-action-token prediction, a diffusion-policy-based head is used, which predicts noise using denoising diffusion probabilistic models (DDPMs). TinyVLA-H, the largest version of TinyVLA, has been found to be 20 times faster than OpenVLA while having 5.5 times fewer parameters [137].

Flow matching is another generative modeling technique used to denoise samples to generate smooth robotic actions for VLAs. π_0 [70] and $\pi_{0.5}$ [85], for instance, use flow matching to generate high-frequency action chunks, and they are considered to be one of the most promising VLAs at present.

5.7. Non-Transformer Backbones

The dominant architecture in VLAs is based on the transformer architecture, which leverages a multi-head attention mechanism, but structured state space models (SSMs) have proven to be efficient in terms of inference speed, such as the Mamba LLM used in RoboMamba [62].

6. Open Challenges and Research Gaps

While VLAs are an active area of research, there are still significant challenges remaining, such as generalization, safety, and efficiency [3]. In [6], some of the key challenges faced when using foundation models in robotics include obtaining large internet-scale data for robotic tasks; the high diversity of tasks, robot platforms, and environments; quantifying uncertainty due to LLM hallucinations; safety aspects of the models; and performance in real-time applications. Some key challenges identified in the included studies are represented by the Word Cloud in Figure 9, with the size of the font proportional to the frequency of the challenge appearing in the studies.

VLAs, when used for robotic manipulation or navigation, should factor safety and security into their design. Many robotic applications operate in environments shared with humans, making unsafe actions potentially harmful. In this co-working environment, the robots need to avoid unsafe navigation, as well as manipulation actions, to avoid hurting humans or damaging other machinery in industrial settings. For home-based, eldercare, or medical robots, they also need to take into consideration the environment and operate safely in a manner that is not dangerous. Lastly, in a swarm setting, when multiple robots

are working together, there should be a safety framework in place that reduces the risk of robots damaging other robots while undertaking any tasks.

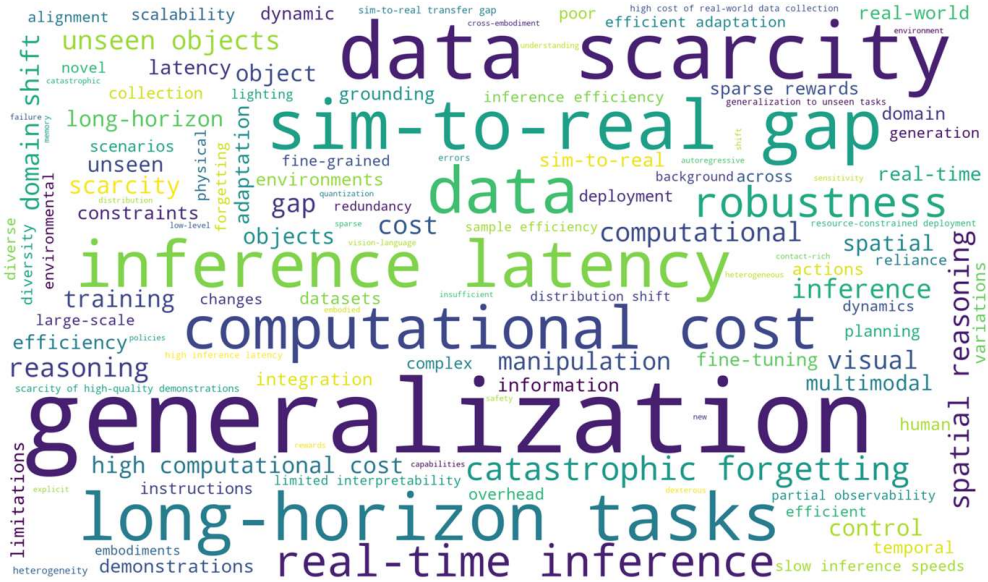


Figure 9. Word cloud of open challenges.

6.1. VLA Security

From a security perspective, VLAs must be safeguarded against adversarial attacks, as vulnerabilities in perception or language grounding can directly induce unsafe physical behavior.

The Physical Vulnerability Evaluation Pipeline (PVEP) proposed by Cheng et al. [191] was used to test LLaRA and OpenVLA models. The main malicious attacks they have been tested against include out-of-distribution (OOD) attacks, typography-based visual prompt attacks, and adversarial patch attacks.

For OOD, where the visual modality was modified by blurring, noise addition, and brightness change, blurring was found to be the most potent, increasing the failure rate to an average of 60.8%. In typography-based visual prompts, text or numbers were injected in a visual modality that increased the failure rate. Finally, adversarial patches were used in four modes, namely black box, robotic black box, gray box, and white box. In black box mode, the attacker does not have any information about the internals of the VLA, and its VLM is used in testing. In the robotic black box, additional robotic data information is provided to generate the adversarial patch. In a gray-box attack, PVEP used VLA and visual question answering (VQA) image prompts for generating the patch. In the white-box attack, all information about VLA and the images and prompts from other tests was used to generate the adversarial patch. In all cases, the failure rate increased, but white-box attacks generated an adversarial patch that increased the failure rate to 94.5%, highlighting the severe implications of such attacks.

In addition, different adversarial attack strategies have been explored by Wang et al. [192], such as Untargeted Action Discrepancy Attack (UADA), Untargeted Position Aware Attack (UPA), and Targeted Manipulation Attack (TMA).

UADA's objective is to maximize the robotic action deviation by introducing a patch that attacks the robot's action space and can therefore lead to a physical hazard. The UPA patch can lead to trajectory distortion by moving the end effector from its intended trajectory. TMA leads the model towards an adversarial target over successive timestamps, thus affecting task success rates.

Another threat to VLAs, called the Greedy Coordinate Gradient (GCG), has been identified in [193], where a suffix text is appended to the text instruction to obtain complete control authority over a VLA. Lastly, backdoor vulnerabilities can be injected during the training stage, which allows VLA models to perform normally in the absence of the visual trigger, but when the trigger is introduced in the perception module of the VLA, it can hijack the policy, as demonstrated in BadVLA [194].

6.2. VLA Safety

Safety is a central concern for VLAs, as they are frequently deployed on robots operating near humans. Robots and humans share the same environment whether the robot is completely independent or working alongside humans. In an industrial robotic environment, there are often emergency stop buttons that can be pressed to stop a robot if its movement is a hazard to human workers or the environment.

Safety can be engineered into the control policy, fine-tuned for specific tasks by introducing human intervention during training, such as in ConRFT, where during online reinforcement-learning-based fine-tuning, a human can intervene using SpaceMouse to ensure safe exploration [29].

VLAs such as SwitchVL [96] can be explored to provide a virtual alternative for stop buttons, where the VLA action can be stopped by issuing a stop instruction in the middle of an unsafe existing instruction execution. A failure in robot task execution can lead to hazardous conditions; thus, multi-task failure detection methods such as SAFE [195] are very important for VLAs. In real-world environments, an embodied agent should not blindly start executing a lingual instruction but should be able to reject it if it is not safe, as incorporated in RationalVLA [98]. SafeVLA [162] demonstrates how to ensure safety by constraining VLA policies through constrained Markov decision process (CMDP)-based safe reinforcement learning (SafeRL).

6.3. High-Computational-Budget Requirements

VLAs have high computational requirements at fine-tuning and inference stages while simultaneously having limited reasoning capability to undertake complex tasks [62]. One of the most challenging aspects is that VLAs can be quite large and may require powerful GPUs [188]. Another significant challenge is that these models require significant compute and memory for training. As an example, the OpenVLA model was trained on 64 A100 GPUs for 14 days. High computational costs at the inference stage mean that deployment on edge and low-cost devices will be challenging. Most robot setups on which VLAs are tested are not low-cost.

6.4. Data Scarcity

While there are web-scale datasets available for training language and vision models, there is a scarcity of large and diverse robotic interaction datasets [49]. Data scarcity, however, can be addressed by building data-efficient models and developing pipelines that use high-fidelity simulation followed by robust techniques for sim-to-real transfer [196].

6.5. Generalization and Robustness

VLA model performance for robotic manipulation tasks can be affected by changes in camera poses and operating conditions, etc. [197]. In a benchmarking study, it was found that there is a significant difference in performance efficiency across different tasks undertaken by different robots using VLA-based policies [198].

VLATest, a VLA testing and evaluation framework presented in [197], has also found that VLAs struggle to locate and manipulate objects correctly in the presence of confounding objects in the scene and differences in camera angles and lighting conditions. They also

confirmed that the accuracy gap between manipulating seen versus unseen objects is significant. VLATest also found that when instructions are paraphrased differently, the model's performance falters. In essence, the robustness of VLA models can be improved.

6.6. Lack of Speech Modality and Auxiliary Information

For generalist robots working in homes or industrial environments, a voice-based interaction between humans and robots is the most convenient form of communication to instruct robots to undertake a task; however, most VLAs are heavily reliant on text-based instructions. Automatic speech recognition (ASR) can be used to convert voice to text and pass this text to VLAs, as is carried out in RoboNurse [158] and Shake-VLA [160]. The ASR error, however, will propagate in VLA. Furthermore, VLAs in their current shape and form cannot understand auxiliary information that can be derived from speech, such as emotion and intonation [159]. VLAS [159] also highlighted the use case where speech recognition, when coupled with understanding the voice-based command, can provide additional information, such as personalized user-specific task understanding.

6.7. Catastrophic Forgetting

Catastrophic forgetting refers to forgetting previously learnt tasks when the model is trained on a new task. A generalist robot continuously requires learning new skills to undertake embodied tasks without forgetting previous ones. Few-shot action incremental learning (FSAIL) is a continual learning approach proposed by Song et al. [121] to mitigate catastrophic forgetting in transformer-based policies; however, as the generalist robots will be expected to undertake more diverse tasks when deployed in real-world scenarios, more techniques will be needed to minimize the risks of catastrophic forgetting.

6.8. Sim-to-Real Gap

Another challenge in VLAs, similar to other robotic policy methods, is the sim-to-real gap. This gap is manifested when robotic policies trained in simulated environments do not perform well when deployed on real robots. The main rationale behind training these policies in a simulated environment is due to the high cost involved in collecting real-world robot datasets. In ReBot [118], a novel real-to-sim-to-real approach has been suggested, where real-world robot trajectories are first replayed in a simulation environment, and then the simulated movements are integrated with real-world background to generate temporally consistent synthetic videos. The performance of VLAs when deployed on real-world robots will improve, as similar methods are devised to reduce the sim-to-real gap.

6.9. Complex Long-Horizon Task Planning and Execution

It is natural to expect humans to give instructions to robots for undertaking multistep sequential tasks for practical applications that require reasoning and planning instead of simple pick-and-drop manipulation tasks. Examples of long-horizon tasks for domestic robots can be laundry folding, dishwashing or cooking a dish. There have been attempts to enable VLAs to undertake long-horizon tasks such as DexVLA [174] that uses embodied curriculum learning but still finds very long-horizon tasks such as folding crumbled shirts challenging. Even leading generalist policies such as Octo could not generalize well with long-horizon tasks [84]. OneTwoVLA [90] uses a hierarchical System-1, System-2 approach with System 2 responsible for task reasoning and planning and System 1 for generating actions but relies on manual annotation for reasoning content. LoHoVLA [94] resolves this by simultaneously generating subtasks and actions but assumes that a subtask can be completed in a single timestamp, which may not hold true for many practical applications. HiberNAC [199] uses a neuro-inspired multi-agent based hierarchical framework to improve success rate and reduce task completion time.

6.10. Research Gaps

Despite growing interest in developing novel architectures and training strategies for vision–language–action (VLA) models, several recurring gaps remain evident in the literature reviewed. The post hoc quality assessment showed that studies generally described model architectures and training procedures more comprehensively than they addressed robustness, reproducibility, and limitations. Robustness evaluation was particularly limited. Only a small proportion of studies examined model performance under realistic variations, such as changes in background, the presence of distractor objects, variations in illumination, or changes in camera viewpoint relative to the object being grasped or manipulated. More systematic robustness testing is therefore needed to determine whether reported performance can be maintained under conditions that differ from those encountered during training. Greater standardization of evaluation protocols is also required. VLA models are frequently assessed using different tasks, datasets, experimental settings, and performance measures, which makes direct comparison difficult. Although task success rate is commonly reported, relatively few studies provide information on inference speed, latency, computational requirements, or resource consumption, despite the importance of these measures for real-time deployment. Furthermore, studies involving physical robots typically rely on expensive research-grade platforms. It therefore remains unclear whether current VLA models can be deployed effectively on lower-cost, consumer-grade hardware with more limited sensing and computational capabilities.

Safety and security are also insufficiently evaluated. Few studies incorporate systematic assessments of operational safety, failure behavior, resilience to unexpected inputs, or security-related risks into their experimental protocols, even though these factors are essential for deployment in real-world environments. Future work should therefore treat safety, security, and failure analysis as integral components of model evaluation rather than as secondary considerations.

Reproducibility represents another important limitation. Greater transparency is needed through the public release of source code, model configurations, training datasets or dataset documentation, evaluation scripts, and sufficiently detailed training recipes. Although results for fixed tasks in simulated environments may be comparatively straightforward to reproduce, replicating experiments involving physical embodied agents can require substantial financial, computational, and infrastructural resources. The adoption of unbiased and comparable evaluation frameworks for generalist robotic policies, such as RoboArena, introduced by Atreya et al. [200], could help address this problem. Reporting VLA performance using standardized frameworks would strengthen the comparability of findings and provide more robust evidence for claims of performance improvement since heterogeneity of benchmarks, robots, tasks, and evaluation protocols prevents a reliable quantitative ranking of different architectures.

Finally, relatively few studies explicitly considered the broader social and practical implications of their work or explained how the reported advances might benefit end users, industry, or society. Future research should more clearly examine the accessibility, affordability, usability, and potential societal impact of VLA systems. Such reporting would strengthen the motivation for the research and clarify the practical relevance and broader applicability of these models.

7. Results

Traditionally, industrial robots have been designed to perform a single repetitive task. However, given the wide range of robotic applications, it is increasingly important to develop methods that can transform robots from specialized single-task machines into generalist, multi-task human co-workers or assistants. This systematic review examined

142 studies to address specific research questions related to the development of robotic policies, the nomenclature used in VLA and robotic learning research, the current state of the art in VLA architectures, strategies for developing efficient VLAs by reducing model size and/or improving inference speed, and the open challenges that remain in this field.

Among the principal approaches for training intelligent robots to undertake different tasks are reinforcement learning, imitation learning and vision–language–action models. These approaches are not mutually exclusive and have also been explored in combination. VLA models may offer certain advantages over alternative methods because they can leverage the knowledge embedded in the underlying vision-language models on which they are built.

Generalist robotic policies are developed using state-of-the-art artificial intelligence algorithms and models, placing this field at the intersection of robotics and AI. Consequently, the literature on robotic policies, including VLA research, draws on a wide range of terminology from both disciplines. Although this nomenclature continues to evolve, it remains important to compile a set of foundational terms, as undertaken in this review.

The current state of the art in VLAs covers a wide variation in architectures and training recipes. VLAs also differ in terms of the modalities they support and the inference optimization methods they employ to reduce the model's size and increase inference speeds. Some VLAs use a hierarchical System-1 and System-2 architecture where System-1 is for high-speed low-level motor control, and System-2 is for reasoning-based task planning. The difference between VLAs could also be due to one of the architectural components, such as a vision encoder, language encoder, fusion mechanism, etc. The three main training approaches are to train the model from scratch using robotic data, fine-tune a VLM and co-training that involves training on both robotic and large-scale visual language datasets. The VLAs that use flow-matching or diffusion-based action policy heads have delivered better performance in general. Spatial information injection can also improve the performance of VLAs for manipulation tasks.

Numerous optimization techniques have been introduced to improve the inference efficiency of VLA models and reduce their computational footprint, with the goal of enabling deployment on physical robots that do not possess high-end GPU resources. Some of the main VLA model optimization methods include quantization, pruning, knowledge distillation, early exit, dynamic layer activation, and action chunking. It is also noteworthy that existing optimization efforts have been concentrated primarily on transformer-based architectures. In contrast, non-transformer alternatives for VLAs, such as state space models (SSMs), remain relatively underexplored.

There are many open challenges related to VLAs that require further research. Most studies included in this review did not cover safety and security aspects of VLAs. VLA security is one of the biggest challenges, where vulnerabilities inherent in the VLMs can be exploited by hackers to hijack VLA policy. Security is always a moving target, so as new security threats emerge for underlying components of VLAs, continuous research is needed to make VLAs resilient to such threats. VLA safety is also of utmost importance to ensure that VLA-based robots operate safely and do not pose any hazard when working around humans and other robots. When it comes to training, the main challenges are high training compute requirements and data scarcity. High computational requirements are also a challenge at the inference stage, as at present, robots cannot use high-end GPUs with sufficient memory to accommodate very large models while remaining cost-effective at the same time.

8. Discussion

The growing interest in VLAs over the past few years highlights their potential as a viable pathway toward generalist robotic policies. This has led researchers to explore diverse architectures, different input modalities, and multiple robotic embodiments. Broadly, VLA-based systems can be divided into monolithic and modular architectures. Monolithic architectures are usually designed around fixed input modalities and predefined VLM and/or LLM components prior to training. By contrast, modular architectures provide greater flexibility, enabling VLAs, VLM backbones, and other components to be replaced, adapted, or combined with relative ease. Across several reviewed studies, diffusion- and flow-matching-based action heads demonstrated improved performance over autoregressive next-token action decoders in specific VLA evaluation settings, suggesting that continuous action generation represents a promising direction for advancing VLA action decoding. For complex long-horizon tasks, hierarchical VLA frameworks with separation of concern between task planning and action execution are a promising direction, with some efforts in combining the two. Hybrid learning-based approaches for improving VLAs through offline and online reinforcement learning are also an active research direction. To make VLAs suitable for deployment in practical applications, efficient model development techniques are improving, relying on model compression and efficient architectural enhancements. It is also evident that there are several useful applications of VLAs that can be beneficial for society, but more research is needed to improve their performance and make them suitable for real-world applications. Future research should explore methods that enable VLAs to perform complex long-horizon tasks, as robotic manipulation systems continue to struggle to achieve acceptable performance in such settings [93]. Further work is also needed to incorporate additional modalities, such as voice and tactile sensing. Improving reasoning capabilities, generalization, and robustness remains an active area of research. In addition, VLAs should be made compatible with a range of robotic embodiments, including low-cost, resource-constrained devices. Automatic recovery from failure is another important capability that generalist robots could leverage when operating in unstructured and dynamic environments. The focus of most of the studies has been on single-arm or bimanual embodiments, with very few studies exploring deploying VLAs on quadruped robots or humanoids, so more research can be beneficial for these embodiments or developing cross-embodiment generalist policies. Finally, further research is needed to ensure that VLA-based generalist agents can operate safely around humans while remaining robust against security vulnerabilities.

9. Limitations

One of the main limitations of this review is that it does not cover relevant publications published after August 2025. In addition, the review includes preprints available during the search period, and these studies may not have undergone peer review at the time of inclusion.

A further methodological limitation is that title and abstract screening, full-text assessment, initial data extraction and post hoc quality assessment were conducted by a single reviewer rather than independently by multiple reviewers. This may have increased the risk of selection and extraction errors. To mitigate this limitation, the extracted data, synthesis, interpretation, and conclusions were reviewed by all authors through multiple rounds of assessment, and any uncertainties were discussed and resolved by consensus.

Another limitation is that the nomenclature in this field is constantly evolving. New terms continue to emerge due to the rapidly developing nature of vision–language–action models and generalist robotic policy research. Therefore, the proposed nomenclature

should ideally be maintained externally through an open-access, web-based portal that can be updated on an ongoing basis by the research community.

Lastly, this review does not include vision–language navigation models, which enable robots to navigate within their environment in response to linguistic commands. Although related to vision-language-action research, these models were considered outside the scope of the present review.

10. Conclusions

Generalist robots capable of performing a wide range of tasks in dynamic and unstructured environments require robust multi-task policies to be effective in their respective application domains. VLAs have emerged as a promising and flexible family of approaches for developing generalist robotic policies. Using the PRISMA methodology, this review examined the state of the art in VLA models, the optimization techniques used to make VLAs more efficient at inference time, the enumeration of robotic policies other than VLAs, and a foundational glossary covering the nomenclature used in VLA and robotic generalist policy research.

Despite substantial progress, important challenges remain. A particularly significant future direction is the integration of the key advances identified in this review toward developing safe, robust, and resilient VLA systems that can run on low-cost hardware, incorporate modalities such as tactile and voice input, and perform a broad range of complex long-range manipulation tasks. Overall, although VLA-based generalist policies represent a promising direction for advancing embodied AI, further research is required to improve their reliability, robustness, safety, scalability, and readiness for practical deployment in diverse real-world environments.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/robotics15080160/s1, Table S1: PRISMA Checklist. Table S2: Search strategies. Table S3: Included Studies, with metadata (submission date, year, authors, title, authors' country of origin, and DOI). Table S4: Code Repositories. Table S5: Quality assessment criteria. Table S6: Post hoc Quality assessment scores of all included studies based on the defined evaluation criteria. The Python script listing is added as a text file in the supplementary material. Reference [201] is cited in the Supplementary Materials.

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Abbreviations

The following abbreviations are used in this manuscript:

AMP	Adversarial Motion Priors
BC	Behavior Cloning
CLIP	Contrastive Language–Image Pre-Training
CMDP	Constrained Markov Decision Process
CoT	Chain-of-Thought
DDPMs	Denoising Diffusion Probabilistic Models
DiT	Diffusion Transformer
DOF	Degree of Freedom
DPO	Direct Preference Optimization
FiLM	Feature-Wise Linear Modulation
FLOPs	Floating-Point Operations
GAI	Generative Artificial Intelligence
GAN	Generative Adversarial Network
GRADE	Grading of Recommendations Assessment, Development and Evaluation
KV	Key–Value
LLM	Large Language Model
LMM	Large Multimodal Model
LoRA	Low-Rank Adaptation
MDP	Markov Decision Process
MLLM	Multimodal Large Language Model
MoE	Mixture of Experts
MPC	Model Predictive Control
PEFT	Parameter-Efficient Fine-Tuning
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
RAG	Retrieval-Augmented Generation
RL	Reinforcement Learning
RPD	Refined Policy Distillation
SLR	Systematic Literature Review
UAV	Unmanned Aerial Vehicle
ViT	Vision Transformer
VLA	Vision-Language-Action
VLM	Vision-Language Model
VLN	Vision-Language-Navigation

Appendix A

Table A1. Glossary of relevant terminology.

Term	Definition
Action Chunking	Grouping of actions that can be executed as one unit is called action chunking. In action chunking, an agent predicts a sequence of actions instead of a single action, and the corresponding policy would therefore become $\pi(a_{t:t+k}\|s_t, o_t; \theta)$ At every timestep, the agent will generate the next k actions [27].
Action Grounding	The process of linking symbolic or high-level natural-language-based instructions to specific perceptual features and objects in the real world is called grounding. Action grounding refers to establishing the connection between natural-language instructions and robot actions. More information available in [203].

Table A1. *Cont.*

Term	Definition
Action Head	Action Head is the final component of VLAs that converts the learned representation of the model into an action for robots.
Adversarial Motion Prior (AMP)	A generative adversarial imitation learning method in which a generative adversarial network (GAN) outputs a trajectory, and the discriminator component learns to distinguish between the trajectory or actions generated by the current policy and the trajectory or actions generated in a particular style (e.g., human-like gait in a robot). See [204].
Affordance	Describes the ability to perform actions on an object in the robot's environment. For example, an affordance can help in assessing object manipulability, such as 'a handle affords grasping'. More information available in [205].
Affordance Grounding	For robots to interact with an object, they need to learn which area of the object the robot can interact with. This task is called affordance grounding. See [206] for more information.
Agent	In robotics, an agent is an entity that perceives its environment through sensors and uses that information and advanced algorithms to decide the actions it takes using actuators. The three capabilities needed for a robot to be termed an intelligent robotic agent are perception, autonomous decision-making, and the ability to take action.
Agentic AI	Artificial intelligence systems that can plan and achieve goals autonomously with minimum human intervention. Most Agentic AI systems at the time of writing this manuscript are based on large language models that can integrate with tools, such as web search, APIs, databases, and code execution engines, to perform a complex multistep task.
Articulated Objects	Objects that consist of multiple links joined together through movable joints, e.g., a laptop, a drawer. See [63].
Attention	Attention refers to a method where neural networks assign different weights, i.e., "attention", to different parts of the input sequence [18]. The idea is to focus on the most important parts of the input sequence.
Behavior Cloning (BC)	A type of imitation learning method where a robot learns a policy through direct supervised learning on state–action data from expert demonstrations.
Bellman Equation	Bellman equation is a concept used in reinforcement learning to find the value function V(s) or Q-function Q(s,a) recursively, where s is the state, and a represents an action. See the state value function and Q-function (also called action-value function) for more information.
Bimanual Manipulation	Bimanual manipulation in robotics refers to robots undertaking tasks using two sets of arms and grippers instead of one. Usually, bimanual manipulation improves reachability and dexterity but comes at the cost of coordination complexity needed to coordinate the movement of both arms to undertake a task [207].
Catastrophic Forgetting	Catastrophic forgetting is a phenomenon in which neural networks abruptly forget the previously learnt tasks when trained on a new task [208]. This is a challenge for continual learning, as we would like robots that can learn new things without forgetting the old ones.
Chain of Thought (CoT)	Chain of thought (CoT) is a prompt engineering technique where a large language model is asked to follow multistep reasoning to improve the output of the model.
Contrastive Language–Image Pre-training (CLIP)	Contrastive Language–Image Pre-training (CLIP) model is a pre-trained model that maps images and text into a shared multimodal embedding space [41]. It is used in robotics for tasks that require language comprehension and visual perception.
Co-fine-tuning	Co-fine-tuning is the process of fine-tuning a pre-trained model on datasets of multiple types. E.g., in [71], a VLM is co-fine-tuned using large visual language and robotic grasping trajectory data.
Configuration Space (C-space)	Configuration in robotics is a complete specification of the position of all points on the robot, and C-space is the space of all possible configurations of a robot. In other words, C-space refers to all possible values that robot joints can take. The minimum number of parameters needed to specify C-space is called the degrees of freedom in robotics.
Contrastive Learning	It is a self-supervised learning method that learns to distinguish similar and dissimilar samples. It brings similar samples closer and dissimilar examples farther in the embedding space. It is used for multimodal alignment, such as aligning text and image embeddings, etc.

Table A1. *Cont.*

Term	Definition
Cross-Attention	Cross-attention enables multiple modalities to attend to each other. As an example, in a vision–language model, cross-attention informs which part/patch of the image is related to the word "apple". Cross-attention is the mechanism whereby one sequence computes an attention score with another sequence.
Curriculum Learning	Curriculum learning is an approach in machine learning where a model is trained using samples in an order whereby easy samples are shown first, followed by the hard samples. The difficulty metric can be defined depending on the modality, e.g., shape complexity for image data. For more information, consult [209].
Dataset Aggregation (DAgger)	DAgger is an imitation learning algorithm that makes IL more robust in the sense that if the robot that is using a policy trained using expert demonstrations through behavior cloning comes across a new unseen state, it asks an expert to recommend the action and adds this state–action pair to the dataset and retrains the policy. At every iteration, DAgger collects a dataset of state–action pairs under the current policy, adds this new data into previously collected data, and retrains the next policy using an aggregate of all data. For more information, consult [210].
Denoising Diffusion Probabilistic Models (DDPMs)	Denoising diffusion probabilistic models are a kind of generative model that are trained to reverse a noising process. In robotics, diffusion policies generate action trajectories. They are known for training stability, the ability to handle high-dimensional action spaces, and their ability to handle multimodal action distributions [17].
Deep Reinforcement Learning	Combines deep learning with reinforcement learning to approximate complex policies or value functions.
Degrees of Freedom (DOFs)	Degrees of Freedom (DOF) of a robot are the number of independent movements a robotic system can make. For a 6-DOF robot arm, 3 are for position, and 3 are for orientation.
Diffusion Policy	Uses diffusion models (that gradually add noise to actions in the forward process and denoise these actions in the reverse process) to generate multistep actions conditioned on sensor data. In robotics, it enables temporally coherent, multimodal action generation. For more information, see [17].
Diffusion Probabilistic Models	See DDPMs.
Direct Preference Optimization	Direct preference optimization (DPO) is an algorithm that directly optimizes the LLMs to align them with human preferences, making them more helpful, harmless, truthful, and safe. They do not need a separate reward model and a separate reinforcement learning loop. For more details, see [211].
Discount Factor	In reinforcement learning, the agent receives a reward for its action to train a policy. The discount factor determines whether the agent prefers immediate (i.e., short-term reward) or future reward. The value of the discount factor ranges from 0 to 1. Usually, it is represented by the letter gamma γ. The following equation shows the discounted return. For more information, consult [212]. $$G_t = \sum_{j=0}^{\infty} \gamma^j R_{t+j+1}$$
Disentanglement	Disentanglement refers to learning a representation of data where variations in an attribute of the data are encoded in a separate axis in lower-dimensional space. It improves the generalization of machine learning algorithms.
Distillation	This is a technique used to transfer knowledge from a large teacher model to a smaller student model. Distillation is also referred to as knowledge distillation and is used to develop efficient policies in robotics to take real-time actions. This is possible since student models are much smaller and faster than the large teacher models.
Dynamics Learning	Dynamics learning is the process through which a robot learns the model that represents the dynamics of the environment. The model can predict the next state given the current state and action.
Dynamic Programming	A collection of algorithms to devise an optimal policy for a given model of the environment.
Dynamic Token Caching	A technique used in LLMs to reduce computational overhead by caching key and value (KV) vectors for repeated tokens, thereby increasing the inference speed.
Egocentric	From the perspective of the robot. As an example, an egocentric image is what the robot sees through its camera.

Table A1. *Cont.*

Term	Definition
Embodied Intelligence	Embodied intelligence is a subfield of artificial intelligence concerned with the interaction between a physical agent and its environment. A robot can learn to walk by interacting with its environment.
Embodied Reasoning	Ability of VLMs to undertake grounding of objects and spatial concepts in the real world [83].
End Effector	The device is attached to the end of a robotic arm, such as a gripper or tool, and it interacts with the environment. Grippers are used for grasping and using objects, such as a parallel jaw gripper, and tools are task-specific end effectors, such as a welding torch in an industrial robot.
Feature-Wise Linear Modulation (FiLM)	Feature-wise linear modulation is used in neural networks to modify intermediate feature maps conditioned on an external input. Used in VLMs to fuse vision and language models. In essence, FiLM makes vision models respond to lingual inputs, thus making them capable of adaptive reasoning. Consult [55] for more information.
Few-Shot Generalization	Few-shot generalization refers to the ability of a model to adapt to new tasks using very few examples. In robotics, few-shot generalization enables robots to handle novel tasks with limited examples available at training time.
Fine-tuning	Fine-tuning is a technique that allows training large pre-trained models using small task-specific datasets for a specific application. In full-fine-tuning, all parameters of the model are updated, whereas in partial-fine-tuning, only the parameters from the last few layers are updated.
Flow Matching	Flow matching is a generative modeling technique that creates a flow to transform a sample taken from a source distribution to a sample from a target distribution. In robotics, flow matching is used to generate smooth high-frequency robot actions by denoising samples from a noise distribution [213].
Forward Kinematics	Helps compute the pose of the robot end effector from the joint angles using transformation matrices. Pose refers to the position and orientation of an object.
Foundation Models	Large-scale models that are trained on very large internet-scale data can subsequently be adapted to undertake many downstream tasks.
Generalist Robot Policy	A single policy that can perform many different tasks under different conditions without requiring retraining.
Gripper	A mechanical end effector used in robotic arms and is designed to grasp and manipulate objects. A parallel jaw is an example of a gripper.
Ground Situation Model (GSM)	A representation of an environment that combines a robot's sensory data with semantic and symbolic information, as well as the spatial and temporal structure of objects, to build actionable world knowledge.
Grounding	Linking symbols and abstract concepts, such as those expressed through words of a language, to real-world sensory input.
Hard Negative Sampling	This is a technique used in contrastive learning that improves model performance by selecting the most challenging negative samples during training to improve model robustness.
Helmholtz Machine	A kind of generative model using recognition (encoder) and generative (decoder) networks to learn the representation of high-dimensional data.
Imitation Learning	Imitation learning is a kind of machine learning in which agents learn a policy by observing and mimicking expert behavior.
In-Context learning	The ability of LLMs to perform tasks by learning from the examples embedded inside the prompt as part of the context.
Inductive Bias	Inductive biases are the assumptions that are built into a learning algorithm that guide generalization. For neural networks, the bias is towards learning non-linear functions.
Inverse Kinematics	Inverse kinematics (IK) allows computing joint angles of a robot from a desired end-effector pose.
Inverse Reinforcement Learning	In IRL, an agent learns the reward function from observed expert behavior. The agent then applies RL to find an optimal policy for that reward.

Table A1. *Cont.*

Term	Definition
Iterative Linear Quadratic Regulator (iLQR)	An optimal control algorithm used in robotics for motion planning, control, and solving trajectory optimization problems by locally approximating dynamics and cost. Examples of iLQR use in robotics include coming up with optimal robot arm trajectories for pick-and-place tasks and planning a flight path for a drone that uses minimum energy.
Kalman Filter	An algorithm to estimate the state of a system from noisy sensory inputs. In robotics, the Kalman filter can be used for navigation, among other applications.
Kinematics	The study of the motion of an object without considering the forces acting upon it. In robotics, kinematics is used to describe the relationships between joint angles and end-effector position.
Kinesthetic Teaching	In kinesthetic teaching, humans teach robots how to undertake a task by physically guiding the robot. A human guide physically moves the end effector of the robot to help it learn how to undertake a specific task.
Kohonen Self-Organizing Network	An unsupervised neural network that projects high-dimensional data to a lower-dimensional space.
Language-assisted Reinforcement Learning (LaRL)	Combines natural language and reinforcement learning to improve generalization. LaRL uses language descriptions or instructions to guide reward shaping or exploration in RL and can improve learning efficiency in robots. The agent learns a language-conditioned policy.
Language-Conditioned Imitation Learning	IL method where the policy is conditioned on natural language input, along with observations from expert demonstrations. Experts perform a task while also providing a textual task description for creating a training dataset. This dataset is used by the robot to learn a language-conditioned policy through behavior cloning, etc.
Language-Conditioned Policy Learning	In language-conditioned policy learning, the agent learns a policy that maps language and observations into an action.
Language-Conditioned Reward Shaping	A technique used in reinforcement learning where the reward function is devised using language inputs. A reward model converts the lingual command into a scalar reward signal, and the agent learns the policy by maximizing the language-conditioned reward. This improves generalization and reduces the effort to develop handcrafted reward signals.
Language–Image Goal-conditioned Value Learning	A technique in reinforcement learning where an agent learns to associate visual observation, language instruction, and a goal state given as an image or lingual description with a value function.
Lifelong Learning	Lifelong learning, also known as continual learning, is a paradigm in machine learning that enables robots to learn continuously without forgetting what they have learnt previously. This capability enables robots to adapt continuously.
Linear Quadratic Regulators (LQRs)	An optimal control algorithm that computes optimal actions minimizing a quadratic cost function under linear dynamics. An example of an application is adjusting rotor speeds of a quadrotor to maintain hover at a location.
Low-Level Control	Low-level control refers to direct control of actuators of a robot, such as joint torques, position, orientation, gripper state, etc.
Low-Rank Adaptation (LoRA)	A parameter-efficient fine-tuning method that helps adapt a large pre-trained model to a downstream task with minimal computation overhead. LoRA injects low-rank trainable matrices into the model's layers, thus reducing the size of trainable parameters.
Markov Decision Process (MDP)	Markov decision process (MDP) is a foundational concept in reinforcement learning that describes a reinforcement learning task that satisfies the Markov property. The Markov property is a constraint that states that the future only depends on the current state and action—thus ignoring history. Mathematically, we can write the Markovian property as follows: $$p(s_{t+1}\|s_t, a_t) = p(s_{t+1}\|s_t, a_t, s_{t-1}, a_{t-1}, \ldots\ldots, s_0, a_0)$$ MDP can be described by a set of states S, a set of actions A, a transition function T, and a reward function R. The dynamics of the system can be represented by the following equation. $$p(s', r\|s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r\|S_t = s, A_t = a\}$$ where s$\in$ S and a$\in$ A.

Table A1. *Cont.*

Term	Definition
Meta-Learning	Meta-learning, often termed as 'Learning to learn,' is a type of machine learning approach where the models learn to generalize across multiple tasks. For instance, robots can quickly adapt to new tasks after seeing only a few examples. For more information, consult [214].
Model Predictive Control (MPC)	MPC is a control strategy that uses a dynamic model of the system to predict future states and plan optimal actions over a future horizon and subsequently replan actions at each timestep. It is used for real-time robotic control.
Model Pruning	A method used to reduce the size of a large model by removing redundant and unimportant components such as weights, neurons, etc., of the model. They improve the inference speed of the models by making them lighter.
Motion Planning	Motion planning is the process of finding a valid sequence of motion for a robot to move from an initial state to a goal state while avoiding obstacles, given the physical constraints on the robot's movement.
Neurosymbolic Reasoning	A method that combines neural networks with symbolic reasoning, such as logic, rules, etc., to develop AI systems that are robust and explainable. In robotics, an example could be combining imitation learning with symbolic safety rules. For more information, consult [215].
Occlusion	Occlusion refers to the partial or complete blockage of an object in the environment from a robot's sensors.
One-Shot Generalization	The ability of artificial intelligence models to generalize and learn something using a single example. In robotics, an example is to learn to undertake a new task by looking at a single expert demonstration or identifying an object by only seeing it once.
Particle Filter	A Monte Carlo localization method is used to determine the location of the robot given the sensor readings. Particles in this case refer to possible locations of the robot with a weight that accounts for the likelihood of that location being the true location [216].
Parallel Decoding	Parallel decoding is a mechanism by virtue of which the VLA generates multiple tokens in parallel instead of predicting tokens through autoregressive decoding. For more information, consult [143].
Parameter Efficient Fine-tuning (PEFT)	A class of fine-tuning algorithms that reduce the number of parameters used in fine-tuning, thus reducing the memory and compute requirements. For more information, see [175].
Policy	A mapping from states or observations to actions that the robot takes. Policy can be deterministic or stochastic in nature.
Partially Observable Markov Decision Process (POMDP)	A partially observable Markov decision process or POMDP is an extension of MDP, where the agent cannot observe the full state.
Pose	Position and orientation of an object or robot in 2D or 3D space.
Proximal Policy Optimization (PPO)	PPO is an algorithm used in reinforcement learning to train optimal policies that prevent overly large policy updates and are sample-efficient. See [217] for more information.
Proprioception	Proprioception refers to the ability of a robot to have internal sensing of its position, movement, and forces by keeping track of joint angles, contact forces of the end effector, etc.
Prompt Engineering	Prompt engineering refers to designing the input prompts to large language models to guide their output. Examples of prompt engineering are few-shot learning and role assignment.
Q-Function	Q-function, also called the action-value function, gives the expected return of taking an action a in the state s as the sum of the immediate reward and the discounted value of the action-value function of the next state–action pair. The Bellman equation for the Q-function is given as follows: $$Q^{\pi}(s,a) = \mathbb{E}_{\pi}[R(s,a) + \gamma Q^{\pi}(S_{t+1}, A_{t+1}) \mid S_t = s, A_t = a]$$ In other words, the Q-function gives the expected return for taking an action a in state s and following the policy π afterwards.
Quantization	Quantization is a model compression technique used in LLMs that reduces the precision of weights and activations from high-precision floating-point data to low-precision data types such as INT4, INT8, etc.

Table A1. *Cont.*

Term	Definition
ReAct	ReAct refers to the reasoning–acting paradigm in which LLMs solve complex tasks by integrating reasoning with action-taking. The actionable steps can interact with external tools such as web search or API calls, while reasoning is verbalized as thoughts. See [218] for more information.
Reasoning Models	Reasoning models are LLMs that are designed to undertake problem-solving in a logically structured fashion to generate a response. They explicitly state their reasoning for arriving at a response.
Reinforcement Learning	Refers to a branch of AI in which an agent learns a policy by maximizing cumulative reward through trial and error.
Reinforcement Learning from Human Feedback (RLHF)	It is a method used to align the output of LLMs with human preferences.
Retrieval-Augmented Generation (RAG)	A technique in which an LLM generates the response using a given context derived from the relevant external data sources.
Reward Function	In reinforcement learning, the reward function maps state–action pairs to scalar values indicating task success. The reward itself is something that impacts the objective function.
Robot Learning	A field at the intersection of machine learning and robotics to enable robots to use perception, decision-making, and control to perform tasks autonomously.
Robotic Foundation Models (RFMs)	RFMs are large, general-purpose models trained on multimodal robotic state–action paired data.
Self-Attention	Self-attention is a mechanism used in transformer-based models that assigns importance scores to each word in a sentence based on its relevance to other words.
Sim2Real	Simulation to reality (Sim2Real) refers to techniques that enable robots trained on simulated data to perform well in the real world. The difference in performance in a simulated environment and the real world is often called the Sim2Real gap.
Simultaneous Localization and Mapping (SLAM)	SLAM is a method that allows a robot to track its own location and build a map of an unknown environment. SLAM typically combines data from multiple sensors such as LiDAR, IMUs, and cameras, with algorithms such as the particle or Kalman filter for state estimation under the hood.
Self-Organizing Map (SOM)	SOM is another term for a Kohonen network. See Kohonen network.
Special Euclidean Groups	Special Euclidean Groups, SE(2) and SE(3), are used to express the pose (i.e., orientation and position) of a robot in 2D and 3D space, respectively. SE(3) can be expressed as (roll, pitch, yaw, x, y, z), and SE(2) can be expressed as (θ, x, y).
Special Orthogonal Groups	Special orthogonal groups, SO(2) and SO(3), are used to express only the orientations in 2D and 3D spaces, respectively.
State Value Function	Estimates the expected return from a state s under a policy π: V(s). The value function guides decision-making in reinforcement learning. The Bellman Equation for the value function is as follows: $$V^{\pi}(s) = \mathbb{E}_{\pi}[R(s,a) + \gamma V^{\pi}(S_{t+1})|S_t = s]$$ Gives the expected cumulative reward of starting from state s and following the policy π afterwards. γ is the discount factor.
Task Planning	Task planning refers to the high-level reasoning to plan a sequence of actions or generate subgoals to achieve a task.
Task Spaces	Task space refers to the set of robot poses that must be reachable by the end effector for a given task.
Teleoperation	Remote control of a robot by a human operator to collect demonstrations or operate in hazardous environments.
Transformer	A transformer is an attention-based deep learning architecture that has revolutionized natural language processing (NLP), generative AI, and robotics through models such as VLAs.

Table A1. *Cont.*

Term	Definition
Vision–Language–Action (VLA)	VLAs are multimodal models that combine visual perception, natural-language-based semantic understanding, and action planning to enable robots to undertake real-world tasks autonomously.
Vision–Language Models (VLM)	VLMs are multimodal models that jointly learn from image and text data. They are used in robotics for task understanding and grounding.
Vision–Language Navigation (VLN)	VLNs use vision and language to navigate environments. A robot with VLNs can follow instructions such as 'go to the dining room'.
Vision Transformer (ViT)	ViT is a transformer model adapted for visual input. Applied in robotic perception for tasks like object recognition, a ViT converts an RGB image into 16 × 16 pixel patches that are projected into patch embeddings [52]
Workspace	In robotics, workspace can be formally defined as the reachable points by the robot. In simple terms, it is the 3D space reachable by the robot's end effector.
World Model	A world model in robotics is a representation of the environment that helps a robot understand how the environment will change in response to its actions and plan optimal actions.
Zero-Shot Generalization	Ability to perform unseen tasks without any task-specific training. Zero-shot generalization is one of the key capabilities expected from generalist robotic agents.

Appendix B

Table A2. Robot setups in VLA research.

Robot Name	Frequency	Type
Franka Emika Panda	42	Robotic Arm
WidowX	23	Robotic Arm
UR5/UR5e	18	Robotic Arm
Google Robot	16	Mobile Robot
Franka Research 3	8	Robotic Arm
AgileX PiPER	7	Robotic Arm
ARX R5	4	Bimanual
Realman RM65-B	4	Robotic Arm
SO-100/101	4	Robotic Arm
UR10/10e	4	Robotic Arm
UR3/3e	4	Robotic Arm
xArm	4	Robotic Arm
Frank Emika Panda dual arm	3	Bimanual
Inspire RH56BFX	3	Dexterous Hand
Sawyer	3	Robotic Arm
AgileX dual-arm	2	Bimanual
ALOHA	2	Bimanual
UR5e dual arm	2	Bimanual
Fourier GR-1 humanoid	2	Humanoid
Kinova Jaco2	2	Robotic Arm
Kuka	2	Robotic Arm

Table A2. *Cont.*

Robot Name	Frequency	Type
AgiBot G-1	1	Humanoid
AgileX Cobot Magic	1	Mobile Robot
ALOHA 2	1	Bimanual
Apptronik Apollo	1	Humanoid
Astribot-S1	1	Humanoid
Barrett Hand BH282	1	Robotic Arm
Baxter	1	Bimanual
Mobile ALOHA	1	Mobile Robot
Continuum robotic endoscope	1	Robotic Endoscope
Elite EC66	1	Robotic Arm
Fair	1	Robotic Arm
FANUC LRMate 200iD/7L	1	Robotic Arm
Flexiv Rizon	1	Robotic Arm
Hello Robot Stretch	1	Mobile Robot
Kinova Gen3	1	Robotic Arm
Mobile ARX	1	Mobile Robot
Mobile Fibocom	1	Mobile Robot
Mobile Trossen	1	Mobile Robot
PsiBot G0-R	1	Dexterous Hand
RealMan RM75-6F	1	Robotic Arm
Spot	1	Quadruped
Trossen Stationary	1	Multiarm Robot
Trossen Viper-X bi-manual robot	1	Bimanual
Unitree G1	1	Humanoid
Unitree Go2	1	Quadruped
Unitree H1	1	Humanoid

References

1. Raiaan, M.A.K.; Mukta, M.d.S.H.; Fatema, K.; Fahad, N.M.; Sakib, S.; Mim, M.M.J.; Ahmad, J.; Ali, M.E.; Azam, S. A Review on Large Language Models: Architectures, Applications, Taxonomies, Open Issues and Challenges. *IEEE Access* **2024**, *12*, 26839–26874. [CrossRef]
2. Pattnayak, P.; Patel, H.L.; Kumar, B.; Agarwal, A.; Banerjee, I.; Panda, S.; Kumar, T. Survey of Large Multimodal Model Datasets, Application Categories and Taxonomy. *arXiv* **2024**, arXiv:2412.17759.
3. Ma, Y.; Song, Z.; Zhuang, Y.; Hao, J.; King, I. A Survey on Vision-Language-Action Models for Embodied AI. *arXiv* **2024**, arXiv:2405.14093.
4. Hu, Y.; Xie, Q.; Jain, V.; Francis, J.; Patrikar, J.; Keetha, N.; Kim, S.; Xie, Y.; Zhang, T.; Fang, H.-S.; et al. Toward General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis. *arXiv* **2024**, arXiv:2312.08782.
5. Yao, X.; Zhou, H.; Mees, O.; Meng, Y.; Xiao, T.; Bisk, Y.; Oh, J.; Johns, E.; Shridhar, M.; Shah, D.; et al. Bridging Language and Action: A Survey of Language-Conditioned Robot Manipulation. *arXiv* **2026**, arXiv:2312.10807.
6. Firoozi, R.; Tucker, J.; Tian, S.; Majumdar, A.; Sun, J.; Liu, W.; Zhu, Y.; Song, S.; Kapoor, A.; Hausman, K.; et al. Foundation Models in Robotics: Applications, Challenges, and the Future. *Int. J. Robot. Res.* **2025**, *44*, 701–739. [CrossRef]
7. Jeong, H.; Lee, H.; Kim, C.; Shin, S. A Survey of Robot Intelligence with Large Language Models. *Appl. Sci.* **2024**, *14*, 8868. [CrossRef]

8. Cui, C.; Ding, P.; Song, W.; Bai, S.; Tong, X.; Ge, Z.; Suo, R.; Zhou, W.; Liu, Y.; Jia, B.; et al. OpenHelix: A Short Survey, Empirical Analysis, and Open-Source Dual-System VLA Model for Robotic Manipulation. *arXiv* **2025**, arXiv:2505.03912.
9. Sapkota, R.; Cao, Y.; Roumeliotis, K.I.; Karkee, M. Vision-Language-Action (VLA) Models: Concepts, Progress, Applications and Challenges. *arXiv* **2026**, arXiv:2505.04769.
10. Din, M.U.; Akram, W.; Saoud, L.S.; Rosell, J.; Hussain, I. Vision Language Action Models in Robotic Manipulation: A Systematic Review. *arXiv* **2025**, arXiv:2507.10672.
11. Zotero. Your Personal Research Assistant. Available online: https://www.zotero.org/ (accessed on 10 February 2025).
12. Jang, E.; Irpan, A.; Khansari, M.; Kappler, D.; Ebert, F.; Lynch, C.; Levine, S.; Finn, C. BC-Z: Zero-Shot Task Generalization with Robotic Imitation Learning. *arXiv* **2022**, arXiv:2202.02005.
13. Lynch, C.; Khansari, M.; Xiao, T.; Kumar, V.; Tompson, J.; Levine, S.; Sermanet, P. Learning Latent Plans from Play. *arXiv* **2019**, arXiv:1903.01973.
14. Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Dabis, J.; Finn, C.; Gopalakrishnan, K.; Hausman, K.; Herzog, A.; Hsu, J.; et al. RT-1: Robotics Transformer for Real-World Control at Scale. *arXiv* **2022**, arXiv:2212.06817.
15. Kalashnikov, D.; Varley, J.; Chebotar, Y.; Swanson, B.; Jonschkowski, R.; Finn, C.; Levine, S.; Hausman, K. MT-Opt: Continuous Multi-Task Robotic Reinforcement Learning at Scale. *arXiv* **2021**, arXiv:2104.08212.
16. Yu, T.; Finn, C.; Xie, A.; Dasari, S.; Zhang, T.; Abbeel, P.; Levine, S. One-Shot Imitation from Observing Humans via Domain-Adaptive Meta-Learning. *arXiv* **2018**, arXiv:1802.01557.
17. Chi, C.; Xu, Z.; Feng, S.; Cousineau, E.; Du, Y.; Burchfiel, B.; Tedrake, R.; Song, S. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. *Int. J. Robot. Res.* **2025**, *44*, 1684–1704. [CrossRef]
18. Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A.N.; Kaiser, Ł.; Polosukhin, I. Attention is All You Need. In *Proceedings of the Advances in Neural Information Processing Systems 30 (NIPS 2017), Long Beach, CA, USA, 4–9 December 2017*; Curran Associates Inc.: Red Hook, NY, USA, 2017; pp. 6000–6010. [CrossRef]
19. Song, C.H.; Wu, J.; Washington, C.; Sadler, B.M.; Chao, W.-L.; Su, Y. LLM-Planner: Few-Shot Grounded Planning for Embodied Agents with Large Language Models. *arXiv* **2023**, arXiv:2212.04088.
20. Suglia, A.; Gao, Q.; Thomason, J.; Thattai, G.; Sukhatme, G. Embodied BERT: A Transformer Model for Embodied, Language-Guided Visual Task Completion. *arXiv* **2021**, arXiv:2108.04927.
21. Ahn, M.; Brohan, A.; Brown, N.; Chebotar, Y.; Cortes, O.; David, B.; Finn, C.; Fu, C.; Gopalakrishnan, K.; Hausman, K.; et al. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. *arXiv* **2022**, arXiv:2204.01691.
22. Zakka, K.; Zeng, A.; Florence, P.; Tompson, J.; Bohg, J.; Dwibedi, D. XIRL: Cross-Embodiment Inverse Reinforcement Learning. *arXiv* **2021**, arXiv:2106.03911.
23. Yu, A.; Mooney, R.J. Using Both Demonstrations and Language Instructions to Efficiently Learn Robotic Tasks. *arXiv* **2023**, arXiv:2210.04476.
24. Reed, S.; Zolna, K.; Parisotto, E.; Colmenarejo, S.G.; Novikov, A.; Barth-Maron, G.; Gimenez, M.; Sulsky, Y.; Kay, J.; Springenberg, J.T.; et al. A Generalist Agent. *arXiv* **2022**, arXiv:2205.06175.
25. Jiang, Y.; Gupta, A.; Zhang, Z.; Wang, G.; Dou, Y.; Chen, Y.; Fei-Fei, L.; Anandkumar, A.; Zhu, Y.; Fan, L. VIMA: Robot Manipulation with Multimodal Prompts. *Proc. Mach. Learn. Res.* **2023**, *202*, 14975–15022.
26. Team, O.M.; Ghosh, D.; Walke, H.; Pertsch, K.; Black, K.; Mees, O.; Dasari, S.; Hejna, J.; Kreiman, T.; Xu, C.; et al. Octo: An Open-Source Generalist Robot Policy. *arXiv* **2024**, arXiv:2405.12213.
27. Zhao, T.Z.; Kumar, V.; Levine, S.; Finn, C. Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware. *arXiv* **2023**, arXiv:2304.13705.
28. Kim, J.W.; Chen, J.-T.; Hansen, P.; Shi, L.X.; Goldenberg, A.; Schmidgall, S.; Scheikl, P.M.; Deguet, A.; White, B.M.; Tsai, D.R.; et al. SRT-H: A Hierarchical Framework for Autonomous Surgery via Language-Conditioned Imitation Learning. *Sci. Robot.* **2025**, *10*, eadt5254. [CrossRef] [PubMed]
29. Chen, Y.; Tian, S.; Liu, S.; Zhou, Y.; Li, H.; Zhao, D. ConRFT: A Reinforced Fine-Tuning Method for VLA Models via Consistency Policy. *arXiv* **2025**, arXiv:2502.05450.
30. Crangle, C.; Suppes, P. Context-Fixing Semantics for Instructable Robots. *Int. J. Man-Mach. Stud.* **1987**, *27*, 371–400. [CrossRef]
31. Maas, R.E.; Suppes, P. Natural-Language Interface for an Instructable Robot. *Int. J. Man-Mach. Stud.* **1985**, *22*, 215–240. [CrossRef]
32. Crangle, C.; Suppes, P. *Language and Learning for Robots*; CSLI Publications: Stanford, CA, USA, 1994.
33. Elshaw, M.; Wermter, S.; Watt, P. Self-Organisation of Language Instruction for Robot Action Control. In *Proceedings of the International Joint Conference on Neural Networks, 2003*; IEEE: Portland, OR, USA, 2003; Volume 1, pp. 22–27.
34. Knoblauch, A.; Markert, H.; Palm, G. An Associative Cortical Model of Language Understanding and Action Planning. In *Artificial Intelligence and Knowledge Engineering Applications: A Bioinspired Approach*; Mira, J., Álvarez, J.R., Eds.; Lecture Notes in Computer Science; Springer: Berlin/Heidelberg, Germany, 2005; Volume 3562, pp. 405–414. [CrossRef]
35. Mavridis, N.; Roy, D. Grounded Situation Models for Robots: Where Words and Percepts Meet. In Proceedings of the 2006 IEEE/RSJ International Conference on Intelligent Robots and Systems, Beijing, China, 9–15 October 2006; pp. 4690–4697.

36. Wermter, S.; Weber, C.; Elshaw, M.; Gallese, V.; Pulvermüller, F. Grounding Neural Robot Language in Action. In *Biomimetic Neural Learning for Intelligent Robots: Intelligent Systems, Cognitive Robotics, and Neuroscience*; Wermter, S., Palm, G., Elshaw, M., Eds.; Springer: Berlin/Heidelberg, Germany, 2005; pp. 162–181.
37. Rickert, M.; Foster, M.E.; Giuliani, M.; By, T.; Panin, G.; Knoll, A. Integrating Language, Vision and Action for Human Robot Dialog Systems. In *Proceedings of the Universal Access in Human-Computer Interaction. Ambient Interaction*; Stephanidis, C., Ed.; Springer: Berlin/Heidelberg, Germany, 2007; pp. 987–995.
38. Brick, T.; Schermerhorn, P.; Scheutz, M. Speech and Action: Integration of Action and Language for Mobile Robots. In Proceedings of the 2007 IEEE/RSJ International Conference on Intelligent Robots and Systems, San Diego, CA, USA, 29 October–2 November 2007; pp. 1423–1428. [CrossRef]
39. Kemke, C. "From Saying to Doing"—Natural Language Interaction with Artificial Agents and Robots. In *Human Robot Interaction*; Sarkar, N., Ed.; IntechOpen: London, UK, 2007; Chapter 9. [CrossRef] [PubMed]
40. Singh, I.; Blukis, V.; Mousavian, A.; Goyal, A.; Xu, D.; Tremblay, J.; Fox, D.; Thomason, J.; Garg, A. ProgPrompt: Program Generation for Situated Robot Task Planning Using Large Language Models. *Auton. Robots* **2023**, *47*, 999–1012. [CrossRef]
41. Radford, A.; Kim, J.W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; et al. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning, Virtual Event, 18–24 July 2021*; Meila, M., Zhang, T., Eds.; Proceedings of Machine Learning Research; PMLR: Cambridge, MA, USA, 2021; Volume 139, pp. 8748–8763.
42. Shah, D.; Osiński, B.; Ichter, B.; Levine, S. LM-Nav: Robotic Navigation with Large Pre-Trained Models of Language, Vision, and Action. *Proc. Mach. Learn. Res.* **2023**, *205*, 492–504.
43. Shridhar, M.; Manuelli, L.; Fox, D. CLIPORT: What and Where Pathways for Robotic Manipulation. *Proc. Mach. Learn. Res.* **2021**, *164*, 894–906.
44. Yuan, L.; Chen, D.; Chen, Y.-L.; Codella, N.C.F.; Dai, X.; Gao, J.; Hu, H.; Huang, X.; Li, B.; Li, C.; et al. Florence: A New Foundation Model for Computer Vision. *arXiv* **2021**, arXiv:2111.11432.
45. Xu, K.; Zhao, S.; Zhou, Z.; Li, Z.; Pi, H.; Wang, Y.; Xiong, R. A Joint Modeling of Vision-Language-Action for Target-Oriented Grasping in Clutter. *arXiv* **2024**, arXiv:2302.12610.
46. Li, X.; Zhang, M.; Geng, Y.; Geng, H.; Long, Y.; Shen, Y.; Zhang, R.; Liu, J.; Dong, H. ManipLLM: Embodied Multimodal Large Language Model for Object-Centric Robotic Manipulation. *arXiv* **2023**, arXiv:2312.16217.
47. Bousmalis, K.; Vezzani, G.; Rao, D.; Devin, C.; Lee, A.X.; Bauza, M.; Davchev, T.; Zhou, Y.; Gupta, A.; Raju, A.; et al. RoboCat: A Self-Improving Generalist Agent for Robotic Manipulation. *arXiv* **2023**, arXiv:2306.11706.
48. Brohan, A.; Brown, N.; Carbajal, J.; Chebotar, Y.; Chen, X.; Choromanski, K.; Ding, T.; Driess, D.; Dubey, A.; Finn, C.; et al. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In *Proceedings of the 7th Conference on Robot Learning, Atlanta, GA, USA, 6–9 November 2023*; Tan, J., Toussaint, M., Darvish, K., Eds.; Proceedings of Machine Learning Research; PMLR: Cambridge, MA, USA, 2023; Volume 229, pp. 2165–2183.
49. O'Neill, A.; Rehman, A.; Maddukuri, A.; Gupta, A.; Padalkar, A.; Lee, A.; Pooley, A.; Gupta, A.; Mandlekar, A.; Jain, A.; et al. Open X-Embodiment: Robotic Learning Datasets and RT-X Models: Open X-Embodiment Collaboration[0]. In Proceedings of the 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 13–17 May 2024; pp. 6892–6903. [CrossRef]
50. Belkhale, S.; Ding, T.; Xiao, T.; Sermanet, P.; Vuong, Q.; Tompson, J.; Chebotar, Y.; Dwibedi, D.; Sadigh, D. RT-H: Action Hierarchies Using Language. In Proceedings of the Robotics: Science and Systems XX, Delft, The Netherlands, 15–19 July 2024. [CrossRef]
51. Kim, M.J.; Pertsch, K.; Karamcheti, S.; Xiao, T.; Balakrishna, A.; Nair, S.; Rafailov, R.; Foster, E.; Lam, G.; Sanketi, P.; et al. OpenVLA: An Open-Source Vision-Language-Action Model. *arXiv* **2024**, arXiv:2406.09246.
52. Dosovitskiy, A.; Beyer, L.; Kolesnikov, A.; Weissenborn, D.; Zhai, X.; Unterthiner, T.; Dehghani, M.; Minderer, M.; Heigold, G.; Gelly, S.; et al. An Image Is Worth 16 × 16 Words: Transformers for Image Recognition at Scale. *arXiv* **2021**, arXiv:2010.11929.
53. Zhai, X.; Mustafa, B.; Kolesnikov, A.; Beyer, L. Sigmoid Loss for Language Image Pre-Training. In *Proceedings of the 2023 IEEE/CVF International Conference on Computer Vision (ICCV)*; IEEE: Paris, France, 2023; pp. 11941–11952.
54. Wang, Y.; Zhu, H.; Liu, M.; Yang, J.; Fang, H.-S.; He, T. VQ-VLA: Improving Vision-Language-Action Models via Scaling Vector-Quantized Action Tokenizers. *arXiv* **2025**, arXiv:2507.01016.
55. Perez, E.; Strub, F.; de Vries, H.; Dumoulin, V.; Courville, A. FiLM: Visual Reasoning with a General Conditioning Layer. *arXiv* **2017**, arXiv:1709.07871.
56. Peebles, W.; Xie, S. Scalable Diffusion Models with Transformers. *arXiv* **2023**, arXiv:2212.09748.
57. Li, H.; Yang, S.; Chen, Y.; Chen, X.; Yang, X.; Tian, Y.; Wang, H.; Wang, T.; Lin, D.; Zhao, F.; et al. CronusVLA: Towards Efficient and Robust Manipulation via Multi-Frame Vision-Language-Action Modeling. *arXiv* **2025**, arXiv:2506.19816.
58. Huang, J.; Yong, S.; Ma, X.; Linghu, X.; Li, P.; Wang, Y.; Li, Q.; Zhu, S.-C.; Jia, B.; Huang, S. An Embodied Generalist Agent in 3D World. *Proc. Mach. Learn. Res.* **2024**, *235*, 20413–20451.

59. Zhen, H.; Qiu, X.; Chen, P.; Yang, J.; Yan, X.; Du, Y.; Hong, Y.; Gan, C. 3D-VLA: A 3D Vision-Language-Action Generative World Model. *Proc. Mach. Learn. Res.* **2024**, *235*, 61229–61245.
60. Gbagbe, K.F.; Cabrera, M.A.; Alabbas, A.; Alyunes, O.; Lykov, A.; Tsetserukou, D. Bi-VLA: Vision-Language-Action Model-Based System for Bimanual Robotic Dexterous Manipulations. In *Proceedings of the 2024 IEEE International Conference on Systems, Man, and Cybernetics (SMC), Kuching, Malaysia, 6–10 October 2024*; IEEE: Piscataway, NJ, USA, 2024; pp. 2864–2869. [CrossRef]
61. Liu, J.; Li, C.; Wang, G.; Lee, L.; Zhou, K.; Chen, S.; Xiong, C.; Ge, J.; Zhang, R.; Zhang, S. Self-Corrected Multimodal Large Language Model for End-to-End Robot Manipulation. *arXiv* **2024**, arXiv:2405.17418.
62. An, P.; Guo, Y.; Li, X.; Liu, J.; Liu, M.; Wang, Z.; Yang, S.; Zhang, R.; Zhang, S.; Zhou, K. RoboMamba: Efficient Vision-Language-Action Model for Robotic Reasoning and Manipulation. In *Proceedings of the Advances in Neural Information Processing Systems 37*; Neural Information Processing Systems Foundation, Inc. (NeurIPS): Vancouver, BC, Canada, 2024; pp. 40085–40110.
63. Huang, S.; Chang, H.; Liu, Y.; Zhu, Y.; Dong, H.; Gao, P.; Boularias, A.; Li, H. A3VLM: Actionable Articulation-Aware Vision Language Model. *arXiv* **2024**, arXiv:2406.07549.
64. Xu, X.; Zhang, Y.; Li, Y.-L.; Han, L.; Lu, C. HumanVLA: Towards Vision-Language Directed Object Rearrangement by Physical Humanoid. *arXiv* **2024**, arXiv:2406.19972.
65. Li, X.; Mata, C.; Park, J.; Kahatapitiya, K.; Jang, Y.S.; Shang, J.; Ranasinghe, K.; Burgert, R.; Cai, M.; Lee, Y.J.; et al. LLaRA: Supercharging Robot Learning Data for Vision-Language Policy. *arXiv* **2024**, arXiv:2406.20095.
66. Ma, Y.; Chi, D.; Wu, S.; Liu, Y.; Zhuang, Y.; Hao, J.; King, I. Actra: Optimized Transformer Architecture for Vision-Language-Action Models in Robot Learning. *arXiv* **2024**, arXiv:2408.01147.
67. Zhang, J.; Guo, Y.; Chen, X.; Wang, Y.-J.; Hu, Y.; Shi, C.; Chen, J. HiRT: Enhancing Robotic Control with Hierarchical Robot Transformers. *arXiv* **2024**, arXiv:2410.05273.
68. Pan, C.; Junge, K.; Hughes, J. Vision-Language-Action Model and Diffusion Policy Switching Enables Dexterous Control of an Anthropomorphic Hand. *arXiv* **2024**, arXiv:2410.14022.
69. Han, B.; Kim, J.; Jang, J. A Dual Process VLA: Efficient Robotic Manipulation Leveraging VLM. *arXiv* **2024**, arXiv:2410.15549.
70. Black, K.; Brown, N.; Driess, D.; Esmail, A.; Equi, M.; Finn, C.; Fusai, N.; Groom, L.; Hausman, K.; Ichter, B.; et al. $\pi 0$: A Vision-Language-Action Flow Model for General Robot Control. *arXiv* **2024**, arXiv:2410.24164.
71. Fan, L.; Chen, K.; Xu, Z.; Yuan, M.; Huang, P.; Huang, W. Language Reasoning in Vision-Language-Action Model for Robotic Grasping. In *Proceedings of the 2024 China Automation Congress (CAC), Qingdao, China, 1–3 November 2024*; IEEE: Piscataway, NJ, USA, 2024; pp. 6656–6661. [CrossRef]
72. Kang, G.-C.; Kim, J.; Shim, K.; Lee, J.K.; Zhang, B.-T. CLIP-RT: Learning Language-Conditioned Robotic Policies from Natural Language Supervision. *arXiv* **2024**, arXiv:2411.00508.
73. Li, Q.; Liang, Y.; Wang, Z.; Luo, L.; Chen, X.; Liao, M.; Wei, F.; Deng, Y.; Xu, S.; Zhang, Y.; et al. CogACT: A Foundational Vision-Language-Action Model for Synergizing Cognition and Action in Robotic Manipulation. *arXiv* **2024**, arXiv:2411.19650.
74. Wen, J.; Zhu, M.; Zhu, Y.; Tang, Z.; Li, J.; Zhou, Z.; Li, C.; Liu, X.; Peng, Y.; Shen, C.; et al. Diffusion-VLA: Scaling Robot Foundation Models via Unified Diffusion and Autoregression. *arXiv* **2024**, arXiv:2412.03293.
75. Zhu, M.; Zhu, Y.; Li, J.; Zhou, Z.; Wen, J.; Liu, X.; Shen, C.; Peng, Y.; Feng, F. ObjectVLA: End-to-End Open-World Object Manipulation Without Demonstration. *arXiv* **2025**, arXiv:2502.19250.
76. Zhong, Y.; Huang, X.; Li, R.; Zhang, C.; Chen, Z.; Guan, T.; Zeng, F.; Lui, K.N.; Ye, Y.; Liang, Y.; et al. DexGraspVLA: A Vision-Language-Action Framework Towards General Dexterous Grasping. *arXiv* **2025**, arXiv:2502.20900.
77. Huang, H.; Liu, F.; Fu, L.; Wu, T.; Mukadam, M.; Malik, J.; Goldberg, K.; Abbeel, P. OTTER: A vision-language-action model with text-aware visual feature extraction. In *Proceedings of the 42nd International Conference on Machine Learning, Vancouver, BC, Canada, 13–19 July 2025*; Singh, A., Fazel, M., Hsu, D., Lacoste-Julien, S., Berkenkamp, F., Maharaj, T., Wagstaff, K., Zhu, J., Eds.; Proceedings of Machine Learning Research; PMLR: Cambridge, MA, USA, 2025; Volume 267, pp. 25566–25580.
78. Li, C.; Wen, J.; Peng, Y.; Peng, Y.; Feng, F.; Zhu, Y. PointVLA: Injecting the 3D World into Vision-Language-Action Models. *arXiv* **2025**, arXiv:2503.07511.
79. Zhao, H.; Song, W.; Wang, D.; Tong, X.; Ding, P.; Cheng, X.; Ge, Z. MoRE: Unlocking Scalability in Reinforcement Learning for Quadruped Vision-Language-Action Models. *arXiv* **2025**, arXiv:2503.08007.
80. Wu, Z.; Zhou, Y.; Xu, X.; Wang, Z.; Yan, H. MoManipVLA: Transferring vision-language-action models for general mobile manipulation. In *Proceedings of the 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 11–15 June 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 1714–1723. [CrossRef]
81. Liu, J.; Chen, H.; An, P.; Liu, Z.; Zhang, R.; Gu, C.; Li, X.; Guo, Z.; Chen, S.; Liu, M.; et al. HybridVLA: Collaborative Diffusion and Autoregression in a Unified Vision-Language-Action Model. *arXiv* **2025**, arXiv:2503.10631.
82. Hou, Z.; Zhang, T.; Xiong, Y.; Duan, H.; Pu, H.; Tong, R.; Zhao, C.; Zhu, X.; Qiao, Y.; Dai, J.; et al. Dita: Scaling Diffusion Transformer for Generalist Vision-Language-Action Policy. *arXiv* **2025**, arXiv:2503.19757.
83. Abeyruwan, S.; Ainslie, J.; Alayrac, J.-B.; Arenas, M.G.; Armstrong, T.; Balakrishna, A.; Baruch, R.; Bauza, M.; Blokzijl, M.; Bohez, S.; et al. Gemini Robotics: Bringing AI into the Physical World. *arXiv* **2025**, arXiv:2503.20020.

84. Tcheurekdjian, D.; Klasmeier, J.; Cooney, T.; McCann, C.; Fenstermaker, T. OPAL: Encoding Causal Understanding of Physical Systems for Robot Learning. *arXiv* **2025**, arXiv:2504.06538.
85. Black, K.; Brown, N.; Darpinian, J.; Dhabalia, K.; Driess, D.; Esmail, A.; Equi, M.R.; Finn, C.; Fusai, N.; Galliker, M.Y.; et al. π0.5: A vision-language-action model with open-world generalization. In *Proceedings of the 9th Conference on Robot Learning, Seoul, Korea, 27–30 September 2025*; Lim, J., Song, S., Park, H.-W., Eds.; Proceedings of Machine Learning Research; PMLR: Cambridge, MA, USA, 2025; Volume 305, pp. 17–40.
86. Deng, S.; Yan, M.; Wei, S.; Ma, H.; Yang, Y.; Chen, J.; Zhang, Z.; Yang, T.; Zhang, X.; Cui, H.; et al. GraspVLA: A Grasping Foundation Model Pre-Trained on Billion-Scale Synthetic Action Data. *arXiv* **2025**, arXiv:2505.03233.
87. Zhu, J.; Sun, X.; Zhang, Q.; Liu, M. VLA-Grasp: A Vision-Language-Action Modeling with Cross-Modality Fusion for Task-Oriented Grasping. *Complex Intell. Syst.* **2025**, *11*, 272. [CrossRef]
88. Bu, Q.; Yang, Y.; Cai, J.; Gao, S.; Ren, G.; Yao, M.; Luo, P.; Li, H. UniVLA: Learning to Act Anywhere with Task-Centric Latent Actions. *arXiv* **2025**, arXiv:2505.06111.
89. Zhao, W.; Li, G.; Gong, Z.; Ding, P.; Zhao, H.; Wang, D. Unveiling the Potential of Vision-Language-Action Models with Open-Ended Multimodal Instructions. *arXiv* **2025**, arXiv:2505.11214.
90. Lin, F.; Nai, R.; Hu, Y.; You, J.; Zhao, J.; Gao, Y. OneTwoVLA: A Unified Vision-Language-Action Model with Adaptive Reasoning. *arXiv* **2025**, arXiv:2505.11917.
91. Song, H.; Qu, D.; Yao, Y.; Chen, Q.; Lv, Q.; Tang, Y.; Shi, M.; Ren, G.; Yao, M.; Zhao, B.; et al. Hume: Introducing System-2 Thinking in Visual-Language-Action Model. *arXiv* **2025**, arXiv:2505.21432.
92. Zhou, Z.; Zhu, Y.; Wen, J.; Shen, C.; Xu, Y. ChatVLA-2: Vision-Language-Action Model with Open-World Embodied Reasoning from Pretrained Knowledge. *arXiv* **2025**, arXiv:2505.21906.
93. Yang, Z.; Chen, Y.; Zhou, X.; Yan, J.; Song, D.; Liu, Y.; Li, Y.; Zhang, Y.; Zhou, P.; Chen, H.; et al. Agentic Robot: A Brain-Inspired Framework for Vision-Language-Action Models in Embodied Agents. *arXiv* **2025**, arXiv:2505.23450.
94. Yang, Y.; Sun, J.; Kou, S.; Wang, Y.; Deng, Z. LoHoVLA: A Unified Vision-Language-Action Model for Long-Horizon Embodied Tasks. *arXiv* **2025**, arXiv:2506.00411.
95. Singh, I.; Goyal, A.; Birchfield, S.; Fox, D.; Garg, A.; Blukis, V. OG-VLA: 3D-Aware Vision Language Action Model via Orthographic Image Generation. *arXiv* **2025**, arXiv:2506.01196.
96. Li, M.; Zhao, Z.; Che, Z.; Liao, F.; Wu, K.; Xu, Z.; Ren, P.; Jin, Z.; Liu, N.; Tang, J. SwitchVLA: Execution-Aware Task Switching for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2506.03574.
97. Li, S.; Gao, L.; Wang, J.; Che, C.; Xiao, X.; Cao, J.; Hu, Y.; Karimi, H.R. Information-Theoretic Graph Fusion with Vision-Language-Action Model for Policy Reasoning and Dual Robotic Control. *Inf. Fusion* **2026**, *131*, 104193. [CrossRef]
98. Song, W.; Chen, J.; Li, W.; He, X.; Zhao, H.; Cui, C.; Su, P.D.S.; Tang, F.; Cheng, X.; Wang, D.; et al. RationalVLA: A Rational Vision-Language-Action Model with Dual System. *arXiv* **2025**, arXiv:2506.10826.
99. Li, P.; Wu, Y.; Xi, Z.; Li, W.; Huang, Y.; Zhang, Z.; Chen, Y.; Wang, J.; Zhu, S.-C.; Liu, T.; et al. ControlVLA: Few-Shot Object-Centric Adaptation for Pre-Trained Vision-Language-Action Models. *arXiv* **2025**, arXiv:2506.16211.
100. Wang, Y.; Li, X.; Wang, W.; Zhang, J.; Li, Y.; Chen, Y.; Wang, X.; Zhang, Z. Unified Vision-Language-Action Model. *arXiv* **2025**, arXiv:2506.19850.
101. Zhang, J.; Chen, Y.; Xu, Y.; Huang, Z.; Zhou, Y.; Yuan, Y.-J.; Cai, X.; Huang, G.; Quan, X.; Xu, H.; et al. 4D-VLA: Spatiotemporal Vision-Language-Action Pretraining with Cross-Scene Calibration. *arXiv* **2025**, arXiv:2506.22242.
102. Argus, M.; Bratulic, J.; Masnavi, H.; Velikanov, M.; Heppert, N.; Valada, A.; Brox, T. cVLA: Towards Efficient Camera-Space VLAs. *arXiv* **2025**, arXiv:2507.02190.
103. Liu, Z.; Gu, Y.; Zheng, S.; Xue, X.; Fu, Y. TriVLA: A Triple-System-Based Unified Vision-Language-Action Model for General Robot Control. *arXiv* **2025**, arXiv:2507.01424.
104. Zhang, W.; Liu, H.; Qi, Z.; Wang, Y.; Yu, X.; Zhang, J.; Dong, R.; He, J.; Wang, H.; Zhang, Z.; et al. DreamVLA: A Vision-Language-Action Model Dreamed with Comprehensive World Knowledge. *arXiv* **2025**, arXiv:2507.04447.
105. Yang, R.; Yu, Q.; Wu, Y.; Yan, R.; Li, B.; Cheng, A.-C.; Zou, X.; Fang, Y.; Cheng, X.; Qiu, R.-Z.; et al. EgoVLA: Learning Vision-Language-Action Models from Egocentric Human Videos. *arXiv* **2025**, arXiv:2507.12440.
106. Yang, S.; Li, H.; Chen, Y.; Wang, B.; Tian, Y.; Wang, T.; Wang, H.; Zhao, F.; Liao, Y.; Pang, J. InstructVLA: Vision-Language-Action Instruction Tuning from Understanding to Manipulation. *arXiv* **2025**, arXiv:2507.17520.
107. Bjorck, J.; Castañeda, F.; Cherniadev, N.; Da, X.; Ding, R.; Fan, L.J.; Fang, Y.; Fox, D.; Hu, F.; Huang, S.; et al. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. *arXiv* **2025**, arXiv:2503.14734.
108. Qu, D.; Song, H.; Chen, Q.; Yao, Y.; Ye, X.; Ding, Y.; Wang, Z.; Gu, J.; Zhao, B.; Wang, D.; et al. SpatialVLA: Exploring Spatial Representations for Visual-Language-Action Model. *arXiv* **2025**, arXiv:2501.15830.
109. Zhou, Z.; Zhu, Y.; Zhu, M.; Wen, J.; Liu, N.; Xu, Z.; Meng, W.; Peng, Y.; Shen, C.; Feng, F.; et al. ChatVLA: Unified Multimodal Understanding and Robot Control with Vision-Language-Action Model. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*; Association for Computational Linguistics: Suzhou, China, 2025; pp. 5377–5395.

110. Zheng, R.; Wang, J.; Reed, S.; Bjorck, J.; Fang, Y.; Hu, F.; Jang, J.; Kundalia, K.; Lin, Z.; Magne, L.; et al. FLARE: Robot Learning with Implicit World Modeling. *arXiv* **2025**, arXiv:2505.15659.
111. Lin, T.; Li, G.; Zhong, Y.; Zou, Y.; Du, Y.; Liu, J.; Gu, E.; Zhao, B. Evo-0: Vision-Language-Action Model with Implicit Spatial Understanding. *arXiv* **2025**, arXiv:2507.00416.
112. Zheng, R.; Liang, Y.; Huang, S.; Gao, J.; Daumé, H.; Kolobov, A.; Huang, F.; Yang, J. TraceVLA: Visual Trace Prompting Enhances Spatial-Temporal Awareness for Generalist Robotic Policies. *arXiv* **2025**, arXiv:2412.10345.
113. Sun, Q.; Hong, P.; Pala, T.D.; Toh, V.; Tan, U.-X.; Ghosal, D.; Poria, S. Emma-X: An Embodied Multimodal Action Model with Grounded Chain of Thought and Look-Ahead Spatial Reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*; Che, W., Nabende, J., Shutova, E., Pilehvar, M.T., Eds.; Association for Computational Linguistics: Vienna, Austria, 2025; pp. 14199–14214.
114. Li, J.; Zhu, Y.; Tang, Z.; Wen, J.; Zhu, M.; Liu, X.; Li, C.; Cheng, R.; Peng, Y.; Peng, Y.; et al. CoA-VLA: Improving Vision-Language-Action Models via Visual-Textual Chain-of-Affordance. *arXiv* **2025**, arXiv:2412.20451.
115. Guo, Y.; Zhang, J.; Chen, X.; Ji, X.; Wang, Y.-J.; Hu, Y.; Chen, J. Improving vision-language-action model with online reinforcement learning. In *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, USA, 19–23 May 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 15665–15672. [CrossRef]
116. Xiang, T.-Y.; Jin, A.-Q.; Zhou, X.-H.; Gui, M.-J.; Xie, X.-L.; Liu, S.-Q.; Wang, S.-Y.; Duang, S.-B.; Wang, S.-C.; Lei, Z.; et al. VLA Model-Expert Collaboration for Bi-Directional Manipulation Learning. *arXiv* **2025**, arXiv:2503.04163.
117. Jülg, T.; Burgard, W.; Walter, F. Refined Policy Distillation: From VLA Generalists to RL Experts. *arXiv* **2025**, arXiv:2503.05833.
118. Fang, Y.; Yang, Y.; Zhu, X.; Zheng, K.; Bertasius, G.; Szafir, D.; Ding, M. ReBot: Scaling Robot Learning with Real-to-Sim-to-Real Robotic Video Synthesis. *arXiv* **2025**, arXiv:2503.14526.
119. Zhao, Q.; Lu, Y.; Kim, M.J.; Fu, Z.; Zhang, Z.; Wu, Y.; Li, Z.; Ma, Q.; Han, S.; Finn, C.; et al. CoT-VLA: Visual chain-of-thought reasoning for vision-language-action models. In *Proceedings of the 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 11–15 June 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 1702–1713. [CrossRef]
120. Ramrakhya, R.; Chang, M.; Puig, X.; Desai, R.; Kira, Z.; Mottaghi, R. Grounding Multimodal LLMs to Embodied Agents That Ask for Help with Reinforcement Learning. *arXiv* **2025**, arXiv:2504.00907.
121. Song, M.; Deng, X.; Zhong, G.; Lv, Q.; Wan, J.; Li, Y.; Hao, J.; Guan, W. Few-Shot Vision-Language Action-Incremental Policy Learning. *arXiv* **2025**, arXiv:2504.15517.
122. Fan, C.; Jia, X.; Sun, Y.; Wang, Y.; Wei, J.; Gong, Z.; Zhao, X.; Tomizuka, M.; Yang, X.; Yan, J.; et al. Interleave-VLA: Enhancing Robot Manipulation with Interleaved Image-Text Instructions. *arXiv* **2025**, arXiv:2505.02152.
123. Bhat, V.; Lan, Y.-H.; Krishnamurthy, P.; Karri, R.; Khorrami, F. 3D CAVLA: Leveraging Depth and 3D Context to Generalize Vision Language Action Models for Unseen Tasks. *arXiv* **2025**, arXiv:2505.05800.
124. Zhang, H.; Zhuang, Z.; Zhao, H.; Ding, P.; Lu, H.; Wang, D. ReinboT: Amplifying Robot Visual-Language Manipulation with Reinforcement Learning. *arXiv* **2025**, arXiv:2505.07395.
125. Lu, W.; Ye, M.; Ye, Z.; Tao, R.; Yang, S.; Zhao, B. RoboFAC: A Comprehensive Framework for Robotic Failure Analysis and Correction. *arXiv* **2025**, arXiv:2505.12224.
126. Zhang, J.; Wu, S.; Luo, X.; Wu, H.; Gao, L.; Shen, H.T.; Song, J. InSpire: Vision-Language-Action Models with Intrinsic Spatial Reasoning. *arXiv* **2025**, arXiv:2505.13888.
127. Tan, S.; Dou, K.; Zhao, Y.; Krähenbühl, P. Interactive Post-Training for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2505.17016.
128. Lu, G.; Guo, W.; Zhang, C.; Zhou, Y.; Jiang, H.; Gao, Z.; Tang, Y.; Wang, Z. VLA-RL: Towards Masterful and General Robotic Manipulation with Scalable Reinforcement Learning. *arXiv* **2025**, arXiv:2505.18719.
129. Vo, T.V.; Nguyen, T.Q.; Nguyen, K.M.; Nguyen, D.H.M.; Vu, M.N. ReFineVLA: Reasoning-Aware Teacher-Guided Transfer Fine-Tuning. *arXiv* **2025**, arXiv:2505.19080.
130. Shu, J.; Lin, Z.; Wang, Y. RFTF: Reinforcement Fine-Tuning for Embodied Agents with Temporal Feedback. *arXiv* **2025**, arXiv:2505.19767.
131. Lu, K.; Ma, C.; Hori, C.; Romeres, D. KitchenVLA: Iterative vision-language corrections for robotic execution of human tasks. In Proceedings of the IEEE International Conference on Robotics and Automation Workshop on Safely Leveraging Vision-Language Foundation Models in Robotics: Challenges and Opportunities (SafeVLMs@ICRA), Atlanta, GA, USA, 23 May 2025.
132. Chen, Z.; Niu, R.; Kong, H.; Wang, Q. TGRPO: Fine-Tuning Vision-Language-Action Model via Trajectory-Wise Group Relative Policy Optimization. *arXiv* **2025**, arXiv:2506.08440.
133. Huang, D.; Fang, Z.; Zhang, T.; Li, Y.; Zhao, L.; Xia, C. CO-RFT: Efficient Fine-Tuning of Vision-Language-Action Models through Chunked Offline Reinforcement Learning. *arXiv* **2025**, arXiv:2508.02219.
134. Sridhar, K.; Dutta, S.; Jayaraman, D.; Lee, I. RICL: Adding In-Context Adaptability to Pre-Trained Vision-Language-Action Models. *arXiv* **2025**, arXiv:2508.02062.

135. Dey, S.; Zaech, J.-N.; Nikolov, N.; Van Gool, L.; Paudel, D.P. ReVLA: Reverting Visual Domain Limitation of Robotic Foundation Models. *arXiv* **2024**, arXiv:2409.15250.
136. Sun, Q.; Lin, H.; Fu, Y.; Fu, Y.; Xue, X. Language guided robotic grasping with fine-grained instructions. In *Proceedings of the 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 1–5 October 2023*; IEEE: Piscataway, NJ, USA, 2023; pp. 1319–1326. [CrossRef]
137. Wen, J.; Zhu, Y.; Li, J.; Zhu, M.; Tang, Z.; Wu, K.; Xu, Z.; Liu, N.; Cheng, R.; Shen, C.; et al. TinyVLA: Toward Fast, Data-Efficient Vision-Language-Action Models for Robotic Manipulation. *IEEE Robot. Autom. Lett.* **2024**, *10*, 3988–3995. [CrossRef]
138. Yue, Y.; Wang, Y.; Kang, B.; Han, Y.; Wang, S.; Song, S.; Feng, J.; Huang, G. DeeR-VLA: Dynamic Inference of Multimodal Large Language Models for Efficient Robot Execution. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*; Curran Associates Inc.: Vancouver, BC, Canada, 2024; Volume 37, pp. 56619–56643.
139. Tong, X.; Ding, P.; Fan, Y.; Wang, D.; Zhang, W.; Cui, C.; Sun, M.; Zhao, H.; Zhang, H.; Dang, Y.; et al. Quart-Online: Latency-free multimodal large language model for quadruped robot learning. In *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, GA, USA, 19–23 May 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 9533–9539. [CrossRef]
140. Xu, S.; Wang, Y.; Xia, C.; Zhu, D.; Huang, T.; Xu, C. VLA-Cache: Efficient Vision-Language-Action Manipulation via Adaptive Token Caching. *arXiv* **2025**, arXiv:2502.02175.
141. Kim, M.J.; Finn, C.; Liang, P. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. *arXiv* **2025**, arXiv:2502.19645.
142. Reuss, M.; Zhou, H.; Rühle, M.; Yağmurlu, Ö.E.; Otto, F.; Lioutikov, R. FLOWER: Democratizing Generalist Robot Policies with Efficient Vision-Language-Action Flow Policies. *arXiv* **2025**, arXiv:2509.04996.
143. Song, W.; Chen, J.; Ding, P.; Zhao, H.; Zhao, W.; Zhong, Z.; Ge, Z.; Li, Z.; Wang, D.; Ma, J.; et al. PD-VLA: Accelerating vision-language-action model integrated with action chunking via parallel decoding. In *Proceedings of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, 19–25 October 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 13162–13169. [CrossRef]
144. Zhang, R.; Dong, M.; Zhang, Y.; Heng, L.; Chi, X.; Dai, G.; Du, L.; Wang, D.; Du, Y.; Zhang, S. MoLe-VLA: Dynamic Layer-Skipping Vision Language Action Model via Mixture-of-Layers for Efficient Robot Manipulation. *AAAI* **2026**, *40*, 18764–18772. [CrossRef]
145. Hung, C.-Y.; Sun, Q.; Hong, P.; Zadeh, A.; Li, C.; Tan, U.-X.; Majumder, N.; Poria, S. NORA: A Small Open-Sourced Generalist Vision Language Action Model for Embodied Tasks. *arXiv* **2025**, arXiv:2504.19854.
146. Kwon, O.; George, A.; Bartsch, A.; Farimani, A.B. RT-Cache: Training-Free Retrieval for Real-Time Manipulation. *arXiv* **2025**, arXiv:2505.09040.
147. Tan, X.; Yang, Y.; Ye, P.; Zheng, J.; Bai, B.; Wang, X.; Hao, J.; Chen, T. Think Twice, Act Once: Token-Aware Compression and Action Reuse for Efficient Inference in Vision-Language-Action Models. *arXiv* **2025**, arXiv:2505.21200.
148. Jiang, F.; Zheng, Z.; Cui, X.; Li, M.; Chen, J.; Chen, X. EaqVLA: Encoding-Aligned Quantization for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2505.21567.
149. Park, S.; Kim, H.; Kim, S.; Jeon, W.; Yang, J.; Jeon, B.; Oh, Y.; Choi, J. Saliency-Aware Quantized Imitation Learning for Efficient Robotic Control. *arXiv* **2025**, arXiv:2505.15304.
150. Shukor, M.; Aubakirova, D.; Capuano, F.; Kooijmans, P.; Palma, S.; Zouitine, A.; Aractingi, M.; Pascal, C.; Russi, M.; Marafioti, A.; et al. SmolVLA: A Vision-Language-Action Model for Affordable and Efficient Robotics. *arXiv* **2025**, arXiv:2506.01844.
151. Duan, Z.; Zhang, Y.; Geng, S.; Liu, G.; Boedecker, J.; Lu, C.X. Fast ECoT: Efficient Embodied Chain-of-Thought via Thoughts Reuse. *arXiv* **2025**, arXiv:2506.07639.
152. Wang, H.; Xiong, C.; Wang, R.; Chen, X. BitVLA: 1-Bit Vision-Language-Action Models for Robotics Manipulation. *arXiv* **2025**, arXiv:2506.07530.
153. Li, Y.; Meng, Y.; Sun, Z.; Ji, K.; Tang, C.; Fan, J.; Ma, X.; Xia, S.; Wang, Z.; Zhu, W. SP-VLA: A Joint Model Scheduling and Token Pruning Approach for VLA Model Acceleration. *arXiv* **2025**, arXiv:2506.12723.
154. Chen, Y.; Li, X. RLRC: Reinforcement Learning-Based Recovery for Compressed Vision-Language-Action Models. *arXiv* **2025**, arXiv:2506.17639.
155. Lin, J.; Taherin, A.; Akbari, A.; Akbari, A.; Lu, L.; Chen, G.; Padir, T.; Yang, X.; Chen, W.; Li, Y.; et al. VOTE: Vision-Language-Action Optimization with Trajectory Ensemble Voting. *arXiv* **2025**, arXiv:2507.05116.
156. Hancock, A.J.; Ren, A.Z.; Majumdar, A. Run-Time Observation Interventions Make Vision-Language-Action Models More Visually Robust. *arXiv* **2024**, arXiv:2410.01971.
157. Budzianowski, P.; Maa, W.; Freed, M.; Mo, J.; Hsiao, W.; Xie, A.; Młoduchowski, T.; Tipnis, V.; Bolte, B. EdgeVLA: Efficient Vision-Language-Action Models. *arXiv* **2025**, arXiv:2507.14049.
158. Li, S.; Wang, J.; Dai, R.; Ma, W.; Ng, W.Y.; Hu, Y.; Li, Z. RoboNurse-VLA: Robotic scrub nurse system based on vision-language-action model. In *Proceedings of the 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Hangzhou, China, 19–25 October 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 3986–3993. [CrossRef]

159. Zhao, W.; Ding, P.; Zhang, M.; Gong, Z.; Bai, S.; Zhao, H.; Wang, D. VLAS: Vision-Language-Action Model With Speech Instructions For Customized Robot Manipulation. *arXiv* **2025**, arXiv:2502.13508.
160. Khan, M.H.; Asfaw, S.; Iarchuk, D.; Altamirano Cabrera, M.; Moreno, L.; Tokmurziyev, I.; Tsetserukou, D. Shake-VLA: Vision-language-action model-based system for bimanual robotic manipulations and liquid mixing. In *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI), Melbourne, Australia, 4–6 March 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 1393–1397. [CrossRef]
161. Ding, P.; Ma, J.; Tong, X.; Zou, B.; Luo, X.; Fan, Y.; Wang, T.; Lu, H.; Mo, P.; Liu, J.; et al. Humanoid-VLA: Towards Universal Humanoid Control with Visual Integration. *arXiv* **2025**, arXiv:2502.14795.
162. Zhang, B.; Zhang, Y.; Ji, J.; Lei, Y.; Cai, Y.; Dai, J.; Chen, Y.; Yang, Y. SafeVLA: Towards Safety Alignment of Vision-Language-Action Model via Constrained Learning. *arXiv* **2025**, arXiv:2503.03480.
163. Zhou, X.; Han, X.; Yang, F.; Ma, Y.; Tresp, V.; Knoll, A. OpenDriveVLA: Towards End-to-End Autonomous Driving with Large Vision Language Action Model. *arXiv* **2025**, arXiv:2503.23463.
164. Ng, C.K.; Bai, L.; Wang, G.; Wang, Y.; Gao, H.; Yuan, K.; Jin, C.; Zeng, T.; Ren, H. EndoVLA: Dual-Phase Vision-Language-Action Model for Autonomous Tracking in Endoscopy. *arXiv* **2025**, arXiv:2505.15206.
165. Yang, Z.; Chai, Y.; Jia, X.; Li, Q.; Shao, Y.; Zhu, X.; Su, H.; Yan, J. DriveMoE: Mixture-of-Experts for Vision-Language-Action Model in End-to-End Autonomous Driving. *arXiv* **2025**, arXiv:2505.16278.
166. Yu, J.; Liu, H.; Yu, Q.; Ren, J.; Hao, C.; Ding, H.; Huang, G.; Huang, G.; Song, Y.; Cai, P.; et al. ForceVLA: Enhancing VLA Models with a Force-Aware MoE for Contact-Rich Manipulation. *arXiv* **2025**, arXiv:2505.22159.
167. Li, P.; Chen, Y.; Wu, H.; Ma, X.; Wu, X.; Huang, Y.; Wang, L.; Kong, T.; Tan, T. BridgeVLA: Input-Output Alignment for Efficient 3D Manipulation Learning with Vision-Language Models. *arXiv* **2025**, arXiv:2506.07961.
168. Huang, J.; Wang, S.; Lin, F.; Hu, Y.; Wen, C.; Gao, Y. Tactile-VLA: Unlocking Vision-Language-Action Model's Physical Knowledge for Tactile Generalization. *arXiv* **2025**, arXiv:2507.09160.
169. Bi, J.; Ma, K.Y.; Hao, C.; Shou, M.Z.; Soh, H. VLA-Touch: Enhancing Vision-Language-Action Models with Dual-Level Tactile Feedback. *arXiv* **2025**, arXiv:2507.17294.
170. Miao, C.; Chang, T.; Wu, M.; Xu, H.; Li, C.; Li, M.; Wang, X. FedVLA: Federated vision-language-action learning with dual gating mixture-of-experts for robotic manipulation. In *Proceedings of the 2025 IEEE/CVF International Conference on Computer Vision (ICCV), Honolulu, HI, USA, 19–23 October 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 7691–7701. [CrossRef]
171. Ding, P.; Zhao, H.; Zhang, W.; Song, W.; Zhang, M.; Huang, S.; Yang, N.; Wang, D. QUAR-VLA: Vision-Language-Action Model for Quadruped Robots. *arXiv* **2023**, arXiv:2312.14457.
172. Sautenkov, O.; Yaqoot, Y.; Lykov, A.; Mustafa, M.A.; Tadevosyan, G.; Akhmetkazy, A.; Altamirano Cabrera, M.; Martynov, M.; Karaf, S.; Tsetserukou, D. UAV-VLA: Vision-language-action system for large scale aerial mission generation. In *Proceedings of the 2025 ACM/IEEE International Conference on Human-Robot Interaction (HRI), Melbourne, Australia, 4–6 March 2025*; IEEE Press: Piscataway, NJ, USA, 2025; pp. 1588–1592. [CrossRef]
173. Dong, Z.; Liu, Y.; Li, Y.; Zhao, H.; Hao, J. Conditioning Matters: Training Diffusion Policies Is Faster Than You Think. *arXiv* **2025**, arXiv:2505.11123.
174. Wen, J.; Zhu, Y.; Li, J.; Tang, Z.; Shen, C.; Feng, F. DexVLA: Vision-Language Model with Plug-In Diffusion Expert for General Robot Control. *arXiv* **2025**, arXiv:2502.05855.
175. Xu, L.; Xie, H.; Qin, S.-Z.J.; Tao, X.; Wang, F.L. Parameter-Efficient Fine-Tuning Methods for Pretrained Language Models: A Critical Review and Assessment. *arXiv* **2023**, arXiv:2312.12148.
176. Mittal, M.; Roth, P.; Tigue, J.; Richard, A.; Zhang, O.; Du, P.; Serrano-Muñoz, A.; Yao, X.; Zurbrügg, R.; Rudin, N.; et al. Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning. *arXiv* **2025**, arXiv:2511.04831.
177. ScanBot: Towards Intelligent Surface Scanning in Embodied Robotic Systems. Available online: https://arxiv.org/html/2505.17295v1 (accessed on 29 June 2026).
178. Wen, Y.; Gu, K.; Liu, H.; Zhao, Y.; Wang, T.; Fan, H.; Sun, X. ROSA: Harnessing Robot States for Vision-Language and Action Alignment. *arXiv* **2025**, arXiv:2506.13679.
179. Jones, J.; Mees, O.; Sferrazza, C.; Stachowicz, K.; Abbeel, P.; Levine, S. Beyond Sight: Finetuning Generalist Robot Policies with Heterogeneous Sensors via Language Grounding. *arXiv* **2025**, arXiv:2501.04693.
180. Liu, B.; Zhu, Y.; Gao, C.; Feng, Y.; Liu, Q.; Zhu, Y.; Stone, P. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. *arXiv* **2023**, arXiv:2306.03310.
181. Li, X.; Hsu, K.; Gu, J.; Pertsch, K.; Mees, O.; Walke, H.R.; Fu, C.; Lunawat, I.; Sieh, I.; Kirmani, S.; et al. Evaluating Real-World Robot Manipulation Policies in Simulation. *arXiv* **2024**, arXiv:2405.05941.
182. Mees, O.; Hermann, L.; Rosete-Beas, E.; Burgard, W. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *IEEE Robot. Autom. Lett.* **2022**, *7*, 7327–7334. [CrossRef]
183. Yuan, Y.; Cui, H.; Chen, Y.; Dong, Z.; Ni, F.; Kou, L.; Liu, J.; Li, P.; Zheng, Y.; Hao, J. From Seeing to Doing: Bridging Reasoning and Decision for Robotic Manipulation. *arXiv* **2026**, arXiv:2505.08548.

184. Li, X.; Li, P.; Liu, M.; Wang, D.; Liu, J.; Kang, B.; Ma, X.; Kong, T.; Zhang, H.; Liu, H. Towards Generalist Robot Policies: What Matters in Building Vision-Language-Action Models. *arXiv* **2024**, arXiv:2412.14058.
185. Chen, W.; Belkhale, S.; Mirchandani, S.; Mees, O.; Driess, D.; Pertsch, K.; Levine, S. Training Strategies for Efficient Embodied Reasoning. *arXiv* **2025**, arXiv:2505.08243.
186. Pertsch, K.; Stachowicz, K.; Ichter, B.; Driess, D.; Nair, S.; Vuong, Q.; Mees, O.; Finn, C.; Levine, S. FAST: Efficient Action Tokenization for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2501.09747.
187. Li, Y.; Zhang, T.; Wei, X.; Li, J.; Zhao, L.; Huang, D.; Fang, Z.; Zheng, M.; Dai, W.; He, X. Object-Focus Actor for Data-Efficient Robot Generalization Dexterous Manipulation. *arXiv* **2025**, arXiv:2505.15098.
188. Samson, M.; Muraccioli, B.; Kanehiro, F. Scalable, training-free visual language robotics: A modular multi-model framework for consumer-grade GPUs. In *Proceedings of the 2025 IEEE/SICE International Symposium on System Integration (SII), Munich, Germany, 21–24 January 2025*; IEEE: Piscataway, NJ, USA, 2025; pp. 193–198. [CrossRef]
189. Tunstall, L.; von Werra, L.; Wolf, T. *Natural Language Processing with Transformers: Building Language Applications with Hugging Face*; O'Reilly Media: Sebastopol, CA, USA, 2022.
190. Sadigh, D.; Belkhale, S. MiniVLA: A Better VLA with a Smaller Footprint. Available online: http://ai.stanford.edu/blog/minivla/ (accessed on 7 July 2025).
191. Cheng, H.; Xiao, E.; Wang, Y.; Yu, C.; Sun, M.; Zhang, Q.; Cao, J.; Guo, Y.; Liu, N.; Xu, K.; et al. Manipulation Facing Threats: Evaluating Physical Vulnerabilities in End-to-End Vision Language Action Models. *arXiv* **2024**, arXiv:2409.13174.
192. Wang, T.; Han, C.; Liang, J.C.; Yang, W.; Liu, D.; Zhang, L.X.; Wang, Q.; Luo, J.; Tang, R. Exploring the Adversarial Vulnerabilities of Vision-Language-Action Models in Robotics. *arXiv* **2024**, arXiv:2411.13587.
193. Jones, E.K.; Robey, A.; Zou, A.; Ravichandran, Z.; Pappas, G.J.; Hassani, H.; Fredrikson, M.; Kolter, J.Z. Adversarial Attacks on Robotic Vision Language Action Models. *arXiv* **2025**, arXiv:2506.03350.
194. Zhou, X.; Tie, G.; Zhang, G.; Wang, H.; Zhou, P.; Sun, L. BadVLA: Towards Backdoor Attacks on Vision-Language-Action Models via Objective-Decoupled Optimization. *arXiv* **2025**, arXiv:2505.16640.
195. Gu, Q.; Ju, Y.; Sun, S.; Gilitschenski, I.; Nishimura, H.; Itkina, M.; Shkurti, F. SAFE: Multitask Failure Detection for Vision-Language-Action Models. *arXiv* **2025**, arXiv:2506.09937.
196. Sui, X.; Tian, D.; Sun, Q.; Chen, R.; Choi, D.; Kwok, K.; Poria, S. From Grounding to Manipulation: Case Studies of Foundation Model Integration in Embodied Robotic Systems. *arXiv* **2025**, arXiv:2505.15685.
197. Wang, Z.; Zhou, Z.; Song, J.; Huang, Y.; Shu, Z.; Ma, L. VLATest: Testing and Evaluating Vision-Language-Action Models for Robotic Manipulation. *Proc. ACM Softw. Eng.* **2025**, *2*, 1615–1638. [CrossRef]
198. Guruprasad, P.; Sikka, H.; Song, J.; Wang, Y.; Liang, P.P. Benchmarking Vision, Language, & Action Models on Robotic Learning Tasks. *arXiv* **2024**, arXiv:2411.05821.
199. Wu, H.; Zhang, H.; Zhang, P.; Wang, J.; Wang, C. HiBerNAC: Hierarchical Brain-Emulated Robotic Neural Agent Collective for Disentangling Complex Manipulation. *arXiv* **2025**, arXiv:2506.08296.
200. Atreya, P.; Pertsch, K.; Lee, T.; Kim, M.J.; Jain, A.; Kuramshin, A.; Eppner, C.; Neary, C.; Hu, E.; Ramos, F.; et al. RoboArena: Distributed Real-World Evaluation of Generalist Robot Policies. *arXiv* **2025**, arXiv:2506.18123.
201. Page, M.J.; McKenzie, J.E.; Bossuyt, P.M.; Boutron, I.; Hoffmann, T.C.; Mulrow, C.D.; Shamseer, L.; Tetzlaff, J.M.; Akl, E.A.; Brennan, S.E.; et al. The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ* **2021**, *372*, n71. [CrossRef] [PubMed]
202. Göpfert, J.; Kuckertz, P.; Weinand, J.; Kotzur, L.; Stolten, D. Citation Graph Builder [Computer Software]. GitHub. 2022. Available online: https://github.com/FZJ-IEK3-VSA/citation-graph-builder (accessed on 29 November 2025).
203. Cohen, V.; Liu, J.X.; Mooney, R.; Tellex, S.; Watkins, D. A Survey of Robotic Language Grounding: Tradeoffs between Symbols and Embeddings. *arXiv* **2024**, arXiv:2405.13245.
204. Peng, X.B.; Ma, Z.; Abbeel, P.; Levine, S.; Kanazawa, A. AMP: Adversarial Motion Priors for Stylized Physics-Based Character Control. *ACM Trans. Graph.* **2021**, *40*, 1–20. [CrossRef]
205. Ardón, P.; Pairet, È.; Lohan, K.S.; Ramamoorthy, S.; Petrick, R.P.A. Affordances in Robotic Tasks—A Survey. *arXiv* **2020**, arXiv:2004.07400.
206. Qian, S.; Chen, W.; Bai, M.; Zhou, X.; Tu, Z.; Li, L.E. AffordanceLLM: Grounding affordance from vision language models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Seattle, WA, USA, 17–18 June 2024*; IEEE: Piscataway, NJ, USA, 2024; pp. 7587–7597. [CrossRef]
207. Grotz, M.; Shridhar, M.; Asfour, T.; Fox, D. PerAct2: Benchmarking and Learning for Robotic Bimanual Manipulation Tasks. *arXiv* **2024**, arXiv:2407.00278.
208. Kirkpatrick, J.; Pascanu, R.; Rabinowitz, N.; Veness, J.; Desjardins, G.; Rusu, A.A.; Milan, K.; Quan, J.; Ramalho, T.; Grabska-Barwinska, A.; et al. Overcoming Catastrophic Forgetting in Neural Networks. *Proc. Natl. Acad. Sci. USA* **2017**, *114*, 3521–3526. [CrossRef] [PubMed]
209. Soviany, P.; Ionescu, R.T.; Rota, P.; Sebe, N. Curriculum Learning: A Survey. *arXiv* **2022**, arXiv:2101.10382.

210. Ross, S.; Gordon, G.J.; Bagnell, J.A. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. *arXiv* **2011**, arXiv:1011.0686.
211. Rafailov, R.; Sharma, A.; Mitchell, E.; Ermon, S.; Manning, C.D.; Finn, C. Direct Preference Optimization: Your Language Model Is Secretly a Reward Model. *arXiv* **2024**, arXiv:2305.18290.
212. Sutton, R.S.; Barto, A.G. *Reinforcement Learning: An Introduction*, 2nd ed.; Adaptive Computation and Machine Learning Series; The MIT Press: Cambridge, MA, USA, 2018.
213. Sarrocco, F. Flow Matching Explained: From Noise to Robot Actions. Available online: https://federicosarrocco.com/blog/flow-matching (accessed on 30 November 2025).
214. Vanschoren, J. Meta-Learning: A Survey. *arXiv* **2018**, arXiv:1810.03548.
215. Bhuyan, B.P.; Ramdane-Cherif, A.; Tomar, R.; Singh, T.P. Neuro-Symbolic Artificial Intelligence: A Survey. *Neural Comput. Applic.* **2024**, *36*, 12809–12844. [CrossRef]
216. Interactive Robotics Algorithms. Available online: https://amrl.cs.utexas.edu/interactive-particle-filters/ (accessed on 26 July 2025).
217. Schulman, J.; Wolski, F.; Dhariwal, P.; Radford, A.; Klimov, O. Proximal Policy Optimization Algorithms. *arXiv* **2017**, arXiv:1707.06347.
218. Yao, S.; Zhao, J.; Yu, D.; Du, N.; Shafran, I.; Narasimhan, K.; Cao, Y. ReAct: Synergizing Reasoning and Acting in Language Models. *arXiv* **2022**, arXiv:2210.03629.

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